

HAWK- n Key Recovery Reduces to SVP in Dimension $n/2 + 1$

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Abstract. HAWK is a lattice signature scheme that is currently a third-round candidate in NIST’s post-quantum signature competition. We give an unconditional, deterministic polynomial-time reduction from HAWK- n key recovery over $K_n = \mathbb{Q}(\zeta_{2^\ell})$ to $\text{poly}(n)$ calls to an exact Shortest Vector Problem (SVP) oracle in dimension $n/2 + 1$, where $n = 2^{\ell-1}$ is the ring degree. The reduction uses a nontrivial automorphism of the key lattice, supplied by the Galois involution $\tau : \zeta \mapsto -\zeta$ and recoverable as a shortest vector of a public rank- n lattice isometric, up to scaling, to $\mathbb{Z}^{n/2+1} \oplus \sqrt{2} \mathbb{Z}^{n/2-1}$. Ducas’s block reduction [Duc23; BN24] on this near-hypercubic class finds the automorphism, and the descent of van Gent and Pulles [GP25] recovers the key from it. In the gate-count model of [AGPS20], the attack lowers the key-recovery cost of HAWK-512 from 2^{150} to 2^{108} and of HAWK-1024 from 2^{288} to 2^{182} . We demonstrate this with a practical implementation that recovers a HAWK-256 secret key end-to-end in a few hours on a single server. The construction does not transfer to Falcon. Conductors $m \in \{p^k, 2p^k\}$ (p an odd prime), i.e. the $m > 4$ with cyclic $(\mathbb{Z}/m)^\times$, evade the attack.

Keywords: HAWK · Lattice Isomorphism Problem · Module-LIP · Cyclotomic fields · Key recovery · SVP

1 Introduction

NIST’s post-quantum signature competition [NIST22] attracted several proposals based on hard lattice problems outside the LWE/SIS family. Among these, HAWK [HAWK25], originally proposed in [DPPvW22], bases its security on the module Lattice Isomorphism Problem (module-LIP) over power-of-two cyclotomic rings, and has fast, floating-point-free signing and compact keys. It was recently selected for the third round of evaluation [A1a26], the only lattice-based candidate to advance. The secret key is a short basis $B \in \text{SL}_2(R_n)$ of the standard lattice R_n^2 over $K_n = \mathbb{Q}(\zeta)$, where $\zeta := \zeta_{2^\ell}$ is a primitive 2^ℓ -th root of unity ($\ell \geq 3$) and R_n is its ring of integers. The public key is its Gram matrix $Q = B^* B$ ($*$ is the conjugate-transpose), Hermitian with respect to complex conjugation. Key recovery entails finding any B' with $B'^* B' = Q$ and reduces to the search module-LIP problem (smLIP) for the isometry class of R_n^2 .

The best previously known key-recovery attack is direct Block–Korkine–Zolotarev (BKZ) lattice reduction on the rank- $2n$ key lattice. HAWK’s designers derive blocksizes $\beta_{\text{key}} \in \{211, 452, 940\}$ for the three parameter sets [HAWK25, Table 7] to achieve different security bounds based on best-known attacks. Ducas [Duc23] provides a powerful tool: a block-reduction algorithm that, given a lattice already known to be isometric to a scaled $\mathbb{Z}^d$, returns its hidden orthogonal basis with calls to an exact-SVP oracle in dimension only $\lfloor d/2 \rfloor + 1$. By itself this yields no HAWK attack: applied to the rank- $2n$ key lattice, it gives oracle dimension $n + 1$, with no improvement on the blocksizes above. Nevertheless, NIST’s second-round status report encourages further analysis of HAWK’s “security assumptions, particularly the smLIP problem within the specific structure of cyclotomic number fields” and notes that prior module-LIP techniques “currently appear inapplicable to the complex cyclotomic number fields utilized by HAWK” [Ala26, §3.4].

Prior module-LIP cryptanalysis [GS02; MPPW24; EP24; Che25; LJPW24] works with complex conjugation c , whose fixed field is totally real. We use a second order-2 Galois involution on the base field, $\tau : \zeta \mapsto -\zeta$, whose fixed field is the *complex* cyclotomic subfield $\mathbb{Q}(\zeta^2)$. We show that the associated τ -cocycle $V_\tau = B^{-1}\tau(B)$ is a shortest vector of a publicly computable lattice of rank only n : 2×2 matrices over R_n cut out by $\mathbb{Q}$ -linear constraints with coefficients depending only on Q .

We prove this lattice is isometric to the near-hypercubic lattice $\sqrt{n/4} \mathbb{Z}^{n/2+1} \oplus \sqrt{n/2} \mathbb{Z}^{n/2-1}$ (Proposition 4.5), so Ducas’s block reduction [Duc23; BN24]¹ finds all its shortest vectors with SVP calls in dimension $n/2 + 1$ (Theorem 5.1, Appendix E). Recovering the key from V_τ is then the descent of van Gent and Pulles [GP25]: their kernel sublattice $\{x \in R_n^2 : \tau(x) = V_\tau^{-1}x\}$ is isometric to a scaled $\mathbb{Z}^n$ (Proposition D.2), and one further run of Ducas’s algorithm on it returns an equivalent key. Combined, this is our main result (Theorem 6.1): HAWK key recovery reduces deterministically to $\text{poly}(n)$ calls to exact SVP in dimension $n/2 + 1$. We implement the attack and recover HAWK-256 secret keys end-to-end in §6.2. The construction does not transfer to Falcon (Appendix F). Conductors with cyclic $(\mathbb{Z}/m)^\times$ (for $m > 4$, those of the form p^k or $2p^k$ with p an odd prime) evade it; we expect many other composite conductors do not (§7).

In the gate-count model of the specification [HAWK25, §5.3], the reduction lowers the key-recovery cost of HAWK-512 from 2^{150} to at most 2^{108} gates and of HAWK-1024 from 2^{288} to at most 2^{182} (Table 1, §6.1). §2 outlines the attack; §3 fixes notation; §§4–6 contain the construction, the reduction, and the experiments; §§7–9 cover the scope of the attack, prior work, and open questions.

¹ The term *near-hypercubic* is Bambury and Nguyen’s [BN24]; the precise class meant throughout this paper is fixed in §3.3.

2 Technical Overview

A HAWK public key is a matrix $Q = B^*B \in \text{GL}_2(R_n)$, where $B \in \text{SL}_2(R_n)$ is the secret key, $K_n = \mathbb{Q}(\zeta)$ for a primitive 2^ℓ -th root of unity ζ , $R_n = \mathbb{Z}[\zeta]$, and $n = 2^{\ell-1}$ is the degree of K_n (§3.1). Key recovery asks for any B' with $B'^*B' = Q$; such a B' signs in place of the secret key [HAWK25, §5.1]. The Galois group of $K_n/\mathbb{Q}$ contains three involutions: complex conjugation c , the map $\tau : \zeta \mapsto -\zeta$, and $\sigma := c\tau$ (§3.2). The attack computes the matrix $V_\tau := B^{-1}\tau(B)$ from Q alone (without needing B), then computes a B' from V_τ . It has three steps.

2.1 The attack in three steps

1. Step 1 (§4) computes from Q a basis of a lattice $\Lambda_B^{(\tau)} \subset M_2(R_n)$ of rank n that contains V_τ as a shortest vector in polynomial time.
2. Step 2 (Theorem 5.1) computes all $2(n/2 + 1)$ shortest vectors of $\Lambda_B^{(\tau)}$. §6.1 bounds the number of SVP-oracle calls in dimension $n/2 + 1$ by $O(n^2 \log n)$. In practice, a single progressive sieve sufficed for HAWK-256, §6.2.
3. Step 3 (Proposition D.2, Appendix D) computes B' from the $2(n/2 + 1)$ candidates: exactly two of them, $\pm V_\tau$, satisfy $Y \equiv I \pmod{2}$, and these are followed by at most two further reduction runs, each with SVP-oracle calls in dimension at most $n/2 + 1$.

Theorem 6.1 combines the three steps into a deterministic reduction: for every power of two $n \geq 4$, HAWK- n key recovery reduces to $\text{poly}(n)$ arithmetic operations and $\text{poly}(n)$ calls to an exact-SVP oracle in dimension at most $n/2 + 1$. Table 1 (§6.1) lists the costs per parameter set; Appendix F explains why Falcon is unaffected.

2.2 Derivation overview

Step 1: the cocycle lattice. The attack needs, from the public key alone, a lattice that contains a vector leading back to the secret key. That vector is the cocycle $V_\tau = B^{-1}\tau(B)$. Although it is built from B , the vector satisfies the cocycle equation and an intertwining relation between the two public Gram matrices Q and $\tau(Q)$. Both become $\mathbb{Q}$ -linear conditions on its entries with coefficients computable from the public key (Lemma 4.1). The integer solutions of the two conditions form the lattice $\Lambda_B^{(\tau)}$, of rank n (Lemma 4.3). A $\mathbb{Z}$ -basis of it is computed from Q in polynomial time (§4.1). V_τ is a shortest vector of $\Lambda_B^{(\tau)}$. Lengths are measured by $Q^{(\tau)}(Y) := \text{Tr}_F(\det Y)$, which is the trace from the degree- $n/4$ field F fixed by c and τ , §3.2. $\det V_\tau = 1$ since $\det B = 1$, and so its squared length is the minimum $\text{Tr}_F(1) = n/4$, regardless of the key-generation distribution (Lemma 4.4). The step uses two properties of HAWK: $\det B = 1$, and the public key is the Gram matrix of B .

Step 2: the isometry class and its reduction. Finding the shortest vectors of $\Lambda_B^{(\tau)}$ requires its isometry class. The change of variables $Y \mapsto BY\tau(B)^{-1}$ carries the cocycle lattice of any key isometrically to that of the trivial key $B = I$ (Lemma 4.2). The map is secret, but only its existence matters, so the class can be read off at $B = I$. There, working in the power basis, $(\Lambda_I^{(\tau)}, Q^{(\tau)}) \cong \sqrt{n/4} \mathbb{Z}^{n/2+1} \oplus \sqrt{n/2} \mathbb{Z}^{n/2-1}$ (Proposition 4.5): up to scaling, the cocycle lattice is in the near-hypercubic class $[\mathbb{Z}^p \oplus \sqrt{2} \mathbb{Z}^q]$ with $p = n/2 + 1$, $q = n/2 - 1$.

Ducas’s block-reduction framework [Duc23; BN24] solves LIP for this class with SVP-oracle calls in dimension $p = n/2 + 1$ (Theorem 5.1, proved in Appendix E), returning all $2(n/2 + 1)$ shortest vectors of $\Lambda_B^{(\tau)}$.

Step 3: from a shortest vector to the key. This step is the automorphism framework of [GP25] (§5, Appendix D). Each shortest vector Y defines an order-2 automorphism of the public quadratic module, with fixed sublattice $\mathcal{M}_Y := \{x \in R_n^2 : \tau(x) = Y^{-1}x\}$ under the form $T(x, y) := \text{Tr}_{K_n}(x^*Qy)$. Only $Y = \pm V_\tau$ lead back to the key, and they are the two candidates with $Y \equiv I \pmod{2}$ (the parity test, Appendix D). For $\pm V_\tau$, $(\mathcal{M}_Y, T)$ is isometric to $\sqrt{n} \mathbb{Z}^n$ with the columns of B^{-1} (up to roots of unity) as its shortest vectors (Proposition D.2). One run of the algorithm of [Duc23] returns the shortest vectors with SVP-oracle calls in dimension at most $n/2 + 1$, and any pair u_1, u_2 with $u_1^*Qu_2 = 0$ gives an equivalent key $[u_1 u_2]^{-1}$. Theorem 6.1 assembles the three steps; §6.1 counts the oracle calls and gates.

3 Preliminaries

3.1 Fields, dimensions, and the key-recovery problem

Throughout, $\ell \geq 3$ is fixed, ζ_m denotes a primitive m -th root of unity and $\zeta := \zeta_{2^\ell}$, and

$$K_n = \mathbb{Q}(\zeta), \quad R_n = \mathbb{Z}[\zeta], \\ n := [K_n : \mathbb{Q}] = 2^{\ell-1}, \quad K_n^+ := \mathbb{Q}(\zeta + \zeta^{-1}).$$

Here K_n and R_n are the cyclotomic field and its ring of integers, in the notation of the specification [HAWK25]; n is the ring degree (also the specification’s n), the module lattice R_n^2 has $\mathbb{Z}$ -rank $2n$, K_n^+ is the maximal totally real subfield of K_n , and $[K_n^+ : \mathbb{Q}] = n/2$. For a general number field E , $\mathcal{O}_E$ denotes its ring of integers, and we abbreviate $\text{Tr}_E := \text{Tr}_{E/\mathbb{Q}}$ and $N_E := N_{E/\mathbb{Q}}$ for the trace and norm down to $\mathbb{Q}$. Superscripts on a ring or field carry several meanings, always distinguishable from context:

- the unit group ($R^\times$) and the k -fold Cartesian power (R^k for an integer $k \geq 1$);
- the fixed field or ring of an automorphism, or of the subgroup $\langle \cdots \rangle$ generated by several $(E^\gamma, E^{\langle \tau \rangle}, \mathcal{O}_E^\gamma, R_n^\gamma)$;

- the (-1) -eigenspace of an involution $(E^{\gamma=-1}, R_n^{\tau=-1})$;
- the maximal totally real subfield (E^+) .

For $X \in M_{a \times b}(K_n)$ (the $a \times b$ matrices over K_n ; we abbreviate $M_a := M_{a \times a}$) write $X^* := \overline{X}^T$ where $\bar{\cdot} = c$ is complex conjugation $\zeta \mapsto \zeta^{-1}$; for $x \in K_n$ we set $|x|^2 := x c(x)$, an element of K_n^+ . The HAWK parameter sets HAWK-256, HAWK-512, and HAWK-1024 are $\ell \in \{9, 10, 11\}$, i.e. $n \in \{256, 512, 1024\}$; the names refer to n , as in the specification.

A HAWK secret key is $B \in \text{SL}_2(R_n)$; the public key is $Q = B^* B \in \text{GL}_2(R_n)$, which is Hermitian with $\det Q = 1$. Two keys B, B' are *equivalent* if $B' = UB$ for some $U \in \text{U}_2(R_n) := \{U \in \text{GL}_2(R_n) : U^* U = I\}$ (equivalently $B'^* B' = Q$). The **key-recovery problem** HAWK-KR(n) (smLIP for the isometry class of R_n^2) is: given Q , output any $B' \in \text{GL}_2(R_n)$ with $B'^* B' = Q$. Any such B' suffices to forge signatures [HAWK25, §5.1].

3.2 The three involutions

$\text{Gal}(K_n/\mathbb{Q}) \cong (\mathbb{Z}/2^\ell)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{\ell-2}$ is non-cyclic for $\ell \geq 3$. Its 2-torsion subgroup has order 4, with three nontrivial commuting involutions:

$$c : \zeta \mapsto \zeta^{-1}, \quad \tau : \zeta \mapsto -\zeta, \quad \sigma := c\tau = \tau c : \zeta \mapsto -\zeta^{-1}.$$

Their fixed fields are $K_n^c = K_n^+$ (totally real), $K_n^\tau = \mathbb{Q}(\zeta^2) = \mathbb{Q}(\zeta_{2^{\ell-1}})$ (complex cyclotomic — the field underlying HAWK- $(n/2)$), and K_n^σ (a CM field of degree $n/2$, i.e. a totally imaginary quadratic extension of a totally real field). Set $F := K_n^{(c,\tau)} = K_n^+ \cap K_n^\tau$, totally real of degree $n/4$.

3.3 Lattice conventions

For a Euclidean lattice Λ , $\lambda_1(\Lambda)$ denotes the length of a shortest nonzero vector, $\Lambda^\vee := \{w \in \text{span}(\Lambda) : \langle w, \Lambda \rangle \subseteq \mathbb{Z}\}$ the dual lattice, and $[\Lambda]$ the isometry (LIP) class; between quadratic lattices, $\cong$ always means isometry, never mere $\mathbb{Z}$ -module isomorphism. For $s > 0$, $s \cdot \Lambda$ denotes Λ with all lengths scaled by s (equivalently, its Gram matrix scaled by s^2). We write $A \oplus B$ for the orthogonal direct sum of quadratic lattices, and call a lattice *hypercubic* if it is isometric to a scaling $s \cdot \mathbb{Z}^m$ of the integer lattice for some $s > 0$. Bambury and Nguyen [BN24] call a lattice *near-hypercubic* when its first minimum is almost as short as the integer lattice's, the operational hypothesis being $\lambda_1(\Lambda) \lambda_1(\Lambda^\vee) < 1 - \varepsilon$ for some $\varepsilon = 1/\text{poly}(\text{rank } \Lambda)$ [BN24, Thm 1]. The term names a broad family, and in this paper it always refers to the specific class arising from HAWK, the lattices isometric to a scaling of $\mathbb{Z}^p \oplus \sqrt{2} \mathbb{Z}^q$ (for which $\lambda_1(\Lambda) \lambda_1(\Lambda^\vee) = 1/\sqrt{2}$, for any $p, q \geq 1$ and any scaling).

We equip R_n with the *trace form* $\langle x, y \rangle_{K_n} := \text{Tr}_{K_n}(x c(y))$; the power-of-two trace formula (stated in the proof of Proposition 4.5) makes the power basis

(ζ^j) orthogonal for it, with all squared norms equal to n . The block-reduction notation (primitive sublattices, volumes, projected blocks) used by Algorithm 1, §9, and Appendices D–E is set up at the start of Appendix E.

4 The τ -Cocycle Lattice

The matrix $V_\tau := B^{-1}\tau(B)$ measures the failure of the secret basis to be τ -equivariant. Since $\det B = 1$, its determinant $(\det B)^{-1}\tau(\det B)$ equals 1, and applying τ once more gives the *cocycle equation* $V_\tau \tau(V_\tau) = I$: V_τ is the Galois 1-cocycle of B for the quadratic extension K_n/K_n^τ . On its own this equation pins down nothing: every $A \in \mathrm{GL}_2(K_n)$ yields a solution $A^{-1}\tau(A)$. Two further constraints, both publicly checkable, cut V_τ down to a shortest vector of a rank- n lattice.

4.1 Definition

For $Y \in M_2(K_n)$ write $\mathrm{adj} Y = \begin{pmatrix} Y_{22} & -Y_{12} \\ -Y_{21} & Y_{11} \end{pmatrix}$, so $Y \cdot \mathrm{adj} Y = (\det Y)I$. Define the τ -cocycle lattice of the public key Q , and the quadratic form on it, as

$$\Lambda_B^{(\tau)} := \left\{ Y \in M_2(R_n) : \underbrace{\tau(Y) = \mathrm{adj} Y}_{(C1)} \text{ and } \underbrace{QY = \sigma(Y)^T \tau(Q)}_{(C2)} \right\},$$

$$Q^{(\tau)}(Y) := \mathrm{Tr}_F(\det Y),$$

where Q is the public Gram matrix and $F = K_n^{(c, \tau)}$ the totally real field of degree $n/4$ from §3.2.

Constraint (C1) linearizes the cocycle equation $\tau(Y) = Y^{-1}$ by replacing Y^{-1} with $\mathrm{adj} Y$; constraint (C2) imposes the relation $QY = \sigma(Y)^T \tau(Q)$ between the public Gram matrices Q and $\tau(Q)$ with Y as the unknown.

Both constraints are $\mathbb{Q}$ -linear in the $4n$ rational coordinates of Y (with τ and σ acting entrywise): (C1) has constant coefficients and (C2) has coefficients determined by Q and $\tau(Q)$, both publicly computable. Hence $\Lambda_B^{(\tau)}$ is the integer kernel of a public linear system, and a $\mathbb{Z}$ -basis is computable in polynomial time via Hermite normal form.

The form $Q^{(\tau)}$ is $\mathbb{Q}$ -valued on $\Lambda_B^{(\tau)}$: for $Y \in \Lambda_B^{(\tau)}$, (C1) gives $\tau(\det Y) = \det(\mathrm{adj} Y) = \det Y$, and taking $\det$ of (C2) with $\det Q = \det \tau(Q) = 1$ gives $\sigma(\det Y) = \det Y$, so $\det Y \in K_n^\tau \cap K_n^\sigma = K_n^{(c, \tau)} = F$, hence $\det Y \in \mathcal{O}_F$. (Neither constraint alone suffices: (C1) places $\det Y$ in K_n^τ , (C2) in K_n^σ .) Shortness is measured with respect to $Q^{(\tau)}$ throughout. Explicitly, $\det Y$ is quadratic in the $4n$ rational coordinates of Y , so $Q^{(\tau)}$ is a quadratic form in the usual sense; the associated bilinear form is $(Y, Y') \mapsto \frac{1}{2} \mathrm{Tr}_F(\det(Y+Y') - \det Y - \det Y')$, and the Gram matrices of $\Lambda_B^{(\tau)}$ used elsewhere are taken with respect to it.

4.2 The secret cocycle and the secret-basis isometry

Lemma 4.1. Let $V_\tau := B^{-1}\tau(B) \in \text{SL}_2(R_n)$. Then $V_\tau \in \Lambda_B^{(\tau)}$ and $\det V_\tau = 1$.

Proof. $\det V_\tau = (\det B)^{-1} \cdot \tau(\det B) = 1$. (C1): $\tau(V_\tau) = \tau(B)^{-1}B = V_\tau^{-1} = \text{adj } V_\tau$ (since $\det V_\tau = 1$). (C2): $QV_\tau = B^*B \cdot B^{-1}\tau(B) = c(B)^T \tau(B)$, and

$$\begin{aligned} \sigma(V_\tau)^T \tau(Q) &= [c\tau(B^{-1}) \cdot c(B)]^T \cdot \tau(B^*B) \\ &= [c(B)]^T (c\tau(B))^{-T} \cdot (c\tau(B))^T \tau(B) \\ &= [c(B)]^T \tau(B). \quad \square \end{aligned}$$

The defining constraints involve Q , so the lattice itself depends on the key; the following (secret) isometry shows that its rank, minimum, and isometry class do not.

Lemma 4.2. *The map $\varphi_\tau : Y \mapsto BY\tau(B)^{-1}$ is a $\mathbb{Z}$ -linear bijection $M_2(R_n) \rightarrow M_2(R_n)$ carrying $\Lambda_B^{(\tau)}$ onto $\Lambda_I^{(\tau)}$ (the lattice of §4.1 for the trivial key $B = I$, i.e. $Q = I$) and preserving det (hence $Q^{(\tau)}$).*

Proof. $B, \tau(B) \in \text{SL}_2(R_n)$, so φ_τ is an R_n -bimodule automorphism. For $W = \varphi_\tau(Y)$: $\tau(W) = \tau(B)\tau(Y)B^{-1}$, and $\text{adj } W = \text{adj}(\tau(B)^{-1}) \cdot \text{adj } Y \cdot \text{adj } B = \tau(B) \cdot \text{adj } Y \cdot B^{-1}$ (using $\text{adj } X = X^{-1}$ for $X \in \text{SL}_2$ and $\text{adj}(XYZ) = \text{adj } Z \cdot \text{adj } Y \cdot \text{adj } X$). These agree iff (C1) holds for Y . Rewriting (C2) as $QY\tau(Q)^{-1} = \sigma(Y)^T$ and substituting $Q = c(B)^T B$, $\tau(Q)^{-1} = \tau(B)^{-1}(c\tau)(B)^{-T}$ gives: (C2) $\Leftrightarrow W = \sigma(W)^T$, which is (C2) at $Q = I$. Finally $\det W = \det B \cdot \det Y \cdot \tau(\det B)^{-1} = \det Y$. $\square$

The attacker does not know φ_τ (it depends on B). Its existence shows that $(\Lambda_B^{(\tau)}, Q^{(\tau)})$ and $(\Lambda_I^{(\tau)}, Q^{(\tau)})$ are isometric as quadratic $\mathbb{Z}$ -lattices, so rank, minimum, shortest-vector count, and isometry class can be read off from the $Q = I$ case.

4.3 Rank and explicit parametrization

At $B = I$ the two constraints decouple entrywise, yielding an explicit parametrization and the rank. Write $R_n^\sigma := R_n \cap K_n^\sigma$ for the fixed ring of σ and $R_n^{\tau=-1} := \{x \in R_n : \tau(x) = -x\}$ for the (-1) -eigenspace of τ . We call the intersection of R_n with an eigenspace of an involution an *eigenlattice*.

Lemma 4.3. $\text{rank}_{\mathbb{Z}} \Lambda_B^{(\tau)} = n$, and $\Lambda_I^{(\tau)} = \left\{ \begin{pmatrix} a & b \\ \sigma(b) & \tau(a) \end{pmatrix} : a \in R_n^\sigma, b \in R_n^{\tau=-1} \right\}$.

Proof. By Lemma 4.2 it suffices to parametrize $\Lambda_I^{(\tau)} = \{W \in M_2(R_n) : \tau(W) = \text{adj } W, W = \sigma(W)^T\}$. Writing $W = \begin{pmatrix} a & b \\ b' & a' \end{pmatrix}$, constraint (C1) gives $\tau(a) = a'$, $\tau(b) = -b$, $\tau(b') = -b'$ entrywise, and constraint (C2) gives $a = \sigma(a)$, $a' = \sigma(a')$, $b' = \sigma(b)$. Thus $a \in R_n^\sigma$ is free ($n/2$ $\mathbb{Z}$ -dimensions since $[K_n : K_n^\sigma] = 2$) and $a' = \tau(a)$ is determined, consistently since $\sigma(a') = \sigma\tau(a) = \tau\sigma(a) = a'$. Similarly $b \in R_n^{\tau=-1}$ is free ($n/2$ $\mathbb{Z}$ -dimensions, basis $\{\zeta^{2j+1} : 0 \leq j < n/2\}$) and

$b' = \sigma(b)$ is determined, consistently since $\tau(b') = \tau\sigma(b) = \sigma\tau(b) = -\sigma(b) = -b'$. The total rank is n . $\square$

This computation is where the hypothesis $\tau \neq c$ is used: for $\tau = c$ the composite $\sigma = c\tau$ is the identity, constraint (C2) degenerates, and the analogous lattice has rank $3n/2$ (§8.3, Appendix C).

We write $W(a, b) := \begin{pmatrix} a & b \\ d(b) & \tau(a) \end{pmatrix}$ for the general element of $\Lambda_I^{(\tau)}$, so that $\Lambda_I^{(\tau)} = \{W(a, b) : a \in R_n^\sigma, b \in R_n^{\tau=-1}\}$. Its determinant $a\tau(a) - b\sigma(b)$ simplifies: since $\sigma(a) = a$, $\tau(a) = \tau\sigma(a) = c(a)$ and $a\tau(a) = |a|^2$; since $\tau(b) = -b$, $\sigma(b) = c\tau(b) = -c(b)$ and $-b\sigma(b) = |b|^2$. Hence $\det W(a, b) = |a|^2 + |b|^2$, with no cross term between a and b .

4.4 Positive-definiteness and shortest vectors

Lemma 4.4. (i) $Q^{(\tau)}$ is positive-definite on $\Lambda_B^{(\tau)}$; (ii) $\min_{Y \neq 0} Q^{(\tau)}(Y) = n/4$, attained exactly at the $2(n/2 + 1)$ vectors Y with $Y\tau(Y) = I$ (equivalently $\det Y = 1$).

Proof. By Lemma 4.2 work in $\Lambda_I^{(\tau)}$. By §4.3, $\det W(a, b) = |a|^2 + |b|^2$, a sum of two elements of the form $x c(x)$, totally ≥ 0 in F , and $= 0$ iff $W = 0$. Therefore $Q^{(\tau)}(W) = \text{Tr}_F(|a|^2 + |b|^2) > 0$ for $W \neq 0$.

By the AM–GM inequality over the $[F : \mathbb{Q}] = n/4$ real embeddings, together with $N_F(\det W) \in \mathbb{Z}_{\geq 1}$ (nonzero algebraic integer), we have $Q^{(\tau)}(W) \geq (n/4) N_F(\det W)^{4/n} \geq n/4$. Equality throughout holds iff $\det W = 1$: equality in the AM–GM forces $\det W$ to be constant across the real embeddings of F , and a totally positive constant of integer norm 1 is 1. Conversely every Y with $\det Y = 1$ has $Q^{(\tau)}(Y) = \text{Tr}_F(1) = n/4$. Hence the minimum is $n/4$, attained exactly at the Y with $\det Y = 1$, equivalently (since $Y \cdot \text{adj } Y = (\det Y)I$ and $\text{adj } Y = \tau(Y)$ by (C1)) at the Y with $Y\tau(Y) = I$.

For the count, a shortest vector is a $W(a, b)$ with $|a|^2 + |b|^2 = 1$, both summands totally non-negative algebraic integers in $\mathcal{O}_F$: if both were nonzero, each would lie in $(0, 1)$ at every real embedding, hence have norm in $(0, 1)$, which is impossible. So one summand vanishes, and the nonzero one of a, b is an algebraic integer all of whose archimedean absolute values equal 1, hence a root of unity by Kronecker's theorem. Either $b = 0$ and a is a root of unity of K_n^σ , i.e. $a = \pm 1$ (since $\sigma(\zeta^k) = \zeta^k$ forces $\zeta^{2k} = (-1)^k$, i.e. $k \in \{0, n\}$), giving the 2 vectors $\pm I$; or $a = 0$ and b is a root of unity of K_n with $\tau(b) = -b$, i.e. $b \in \{\zeta^k : k \text{ odd}\}$, giving n vectors. Total $2(n/2 + 1)$. $\square$

4.5 The isometry class

By Lemma 4.2 the isometry class of $(\Lambda_B^{(\tau)}, Q^{(\tau)})$ is that of the trivial-key lattice

Proposition 4.5. $(\Lambda_I^{(\tau)}, Q^{(\tau)}) \cong \sqrt{n/4} \mathbb{Z}^{n/2+1} \oplus \sqrt{n/2} \mathbb{Z}^{n/2-1}$.

Proof. By §4.3, $\Lambda_I^{(\tau)} = \{W(a, b) : a \in R_n^\sigma, b \in R_n^{\tau=-1}\}$ with $\det W(a, b) = |a|^2 + |b|^2$, so the two summands are $Q^{(\tau)}$ -orthogonal and $Q^{(\tau)}(W(a, b)) = \text{Tr}_F(|a|^2) + \text{Tr}_F(|b|^2) = \frac{1}{4}\langle a, a \rangle_{K_n} + \frac{1}{4}\langle b, b \rangle_{K_n}$, where $\langle \cdot, \cdot \rangle_{K_n}$ is the trace form of §3.3 and we used $|a|^2, |b|^2 \in F$ and $[K_n : F] = 4$. It remains to compute the restriction of $\langle \cdot, \cdot \rangle_{K_n}$ to the eigenlattices $R_n^{\tau=-1} = \ker(\tau+1)|_{R_n}$ and $R_n^\sigma = \ker(\sigma-1)|_{R_n}$.

By the power-of-two trace formula ($\text{Tr}_{K_n}(\zeta^m)$ equals n if $m \equiv 0$, $-n$ if $m \equiv n$, else $0 \pmod{2^\ell}$), the power basis $(\zeta^j)_{0 \leq j < n}$ is $\langle \cdot, \cdot \rangle_{K_n}$ -orthogonal with all squared norms equal to n : $(R_n, \langle \cdot, \cdot \rangle_{K_n}) \cong \sqrt{n} \mathbb{Z}^n$. Every $\gamma \in \text{Gal}(K_n/\mathbb{Q})$ is an isometry of this lattice ($\text{Tr}_{K_n}(\gamma(x)\gamma(y)) = \text{Tr}_{K_n}(\gamma(x\bar{y})) = \text{Tr}_{K_n}(x\bar{y})$) and maps each ζ^j to $\pm \zeta^{j'}$, i.e. acts on the orthogonal basis by a signed permutation. For an involution the cycles have length 1 or 2, and the ε -eigenlattice ($\varepsilon = \pm 1$) decomposes orthogonally over the cycles: a basis vector fixed with sign ε contributes itself (squared norm n); a basis vector fixed with sign $-\varepsilon$ contributes nothing; a 2-cycle $\zeta^j \mapsto s\zeta^{j'} \mapsto \zeta^j$ contributes the single vector $\zeta^j + \varepsilon s\zeta^{j'}$ (squared norm $2n$).

For τ : $\tau(\zeta^j) = (-1)^j \zeta^j$ — every basis vector is fixed up to sign, there are no 2-cycles, and the (-1) -eigenlattice is the span of the odd powers: $R_n^{\tau=-1} = \bigoplus_{j \text{ odd}} \mathbb{Z}\zeta^j \cong \sqrt{n} \mathbb{Z}^{n/2}$.

For σ : $\sigma(\zeta^j) = (-1)^j \zeta^{-j} = (-1)^{j+1} \zeta^{n-j}$ for $\boxed{1} \leq \boxed{j \leq n-1}$ (using $\zeta^{-j} = -\zeta^{n-j}$), and $\sigma(1) = 1$. The underlying permutation $j \mapsto n-j$ has two fixed points, $j = 0$ (sign $+1$; this is the case $\sigma(1) = 1$ treated separately above) and $j = n/2$ (sign $(-1)^{n/2+1} = -1$ since $n/2$ is even), and $n/2 - 1$ two-cycles $\{j, n-j\}$; on the rank-2 sublattice $\mathbb{Z}\zeta^j \oplus \mathbb{Z}\zeta^{n-j}$ the involution acts by $\begin{pmatrix} 0 & s \\ s & 0 \end{pmatrix}$ with $s = (-1)^{j+1}$, whose integer fixed vectors are exactly $\mathbb{Z}(\zeta^j + s\zeta^{n-j})$. Hence $R_n^\sigma = \mathbb{Z} \cdot 1 \oplus \bigoplus_{j=1}^{n/2-1} \mathbb{Z}(\zeta^j + (-1)^{j+1} \zeta^{n-j}) \cong \sqrt{n} \mathbb{Z} \oplus \sqrt{2n} \mathbb{Z}^{n/2-1}$.

Dividing all squared norms by 4 ($n/4$ and $n/2$) and combining the two $Q^{(\tau)}$ -orthogonal summands gives the claim. The $n/2 + 1$ short basis vectors (squared norm $n/4$) are $W(1, 0) = I$ and $W(0, \zeta^j)$ for j odd. $\square$

Corollary 4.6. After rescaling by $\sqrt{4/n}$ (i.e. $2/\sqrt{n}$), the public lattice $\Lambda_B^{(\tau)}$ is in the LIP class $[A']$ [DvW21] of the near-hypercubic lattice

$$A' := \mathbb{Z}^p \oplus \sqrt{2} \mathbb{Z}^q, \quad \boxed{p := n/2 + 1, \quad q := n/2 - 1, \quad p + q = n.}$$

The Gram representative $D = \text{diag}(I_p, 2I_q)$ is integral; the dual class is $[\mathbb{Z}^p \oplus \frac{1}{\sqrt{2}} \mathbb{Z}^q]$. (The integers p, q fixed here are used throughout §5 and Appendix E; they are not to be confused with the odd prime p of §7.)

Remark 4.7. The proof of Proposition 4.5 uses $\text{Tr}_{K_n}(\zeta^m) = 0$ for all $0 < |m| < n$. For non-power-of-two conductors this fails (e.g., $\text{Tr}_{\mathbb{Q}(\zeta_p)/\mathbb{Q}}(\zeta_p^m) = -1$ for an odd prime p and all $m \not\equiv 0$), and the analogous lattice need not be isometric to any $a\mathbb{Z}^{p'} \oplus b\mathbb{Z}^{q'}$, $0 < a < b$; see §7.

5 Recovering the Key from $\Lambda_B^{(\tau)}$

Corollary 4.6 places the rescaled cocycle lattice in the LIP class $[A'] = [\mathbb{Z}^p \oplus \sqrt{2} \mathbb{Z}^q]$, $p = n/2 + 1$, $q = n/2 - 1$. We run Ducan's two-block reduction [Duc23, Alg. 1] with cut $k := p - 1$:

Algorithm 1 Ducan's two-block reduction [Duc23, Alg. 1] on $[A']$, cut $k = p - 1$. The reduction operations are defined in §E.4.

Require: integral Gram matrix of a basis $B = (b_0, \dots, b_{p+q-1})$ of $A \in [A']$ (this B is unrelated to the secret key of §3.1)

Ensure: $U \in \text{GL}_{p+q}(\mathbb{Z})$ with U^T (input) $U = \text{diag}(I_p, 2I_q)$ up to signed permutation

- 1: **while** $\text{vol}(B_{[0:k-1]})^2 > 1$ **do**
 - 2: dual-SVP reduce $B_{[0:k]}$; primal-SVP reduce $B_{[k:p+q-1]}$
 - 3: HKZ-reduce $B_{[0:k-1]}$; HKZ-reduce $B_{[k:p+q-1]}$; size-reduce B
-

Ducan's progress lemma [Duc23, Lemma 4] fails on $[A']$ (§E.1); Appendix E adapts the product-progress framework of [BN24, Lemma 1, Thm 1] to $[A']$, supplying the no-abort bound (Lemma E.2; see Remark E.5), and proves:

Theorem 5.1. *On input an integral Gram matrix H of a basis of a lattice in $[A']$, Algorithm 1 terminates after at most $p^2 \ln p + p \ln(2(p-1) \ln \max_i \|b_i\|) + 3p$ iterations (any basis of $[A']$ has $\max_i \|b_i\| \geq \sqrt{2}$) and outputs $U \in \text{GL}_{p+q}(\mathbb{Z})$ with $U^T H U = \text{diag}(I_p, 2I_q)$ up to signed permutation, using SVP-oracle calls in dimension $\leq p = n/2 + 1$ only.*

Proof: Appendix E.

Algorithm 1 returns all $2(n/2 + 1)$ shortest vectors of $\Lambda_B^{(\tau)}$. For each, $x \mapsto Y \tau(x)$ is an order-2 automorphism of the public form (R_n^2, T) outside the trivial group of [GP25] (Lemma D.1), so the descent of van Gent and Pulles [GP25, Props. 1–2] applies: a short vector of the kernel sublattice $\mathcal{M}_Y$ recovers an equivalent key.

Appendix D sharpens the analysis to an unconditional bound: the parity test $Y \equiv I \pmod{2}$ selects the two candidates $Y = \pm V_\tau$, for which the kernel is exactly $\sqrt{n} \mathbb{Z}^n$ (Proposition D.2), so one further run of Ducan's algorithm [Duc23] on it returns the key with SVP-oracle calls in dimension at most $n/2 + 1$.

6 Main Theorem and Security Consequences

Theorem 6.1. *For every power of two $n \geq 4$ there is a deterministic algorithm that, given a HAWK- n public key $Q = B^* B$ with $B \in \text{SL}_2(R_n)$, outputs $B' \in \text{GL}_2(R_n)$ with $B'^* B' = Q$, making $\text{poly}(n)$ arithmetic operations and $\text{poly}(n)$ calls to an exact-SVP oracle in dimension at most $n/2 + 1$.*

Proof.

1. Compute a $\mathbb{Z}$ -basis and Gram matrix $G^{(\tau)}$ of $(\Lambda_B^{(\tau)}, Q^{(\tau)})$ from Q . By [Lemma 4.2](#) and [Proposition 4.5](#), $\frac{4}{n}G^{(\tau)}$ is an integral Gram matrix of a lattice in $[\Lambda']$.
2. LLL-reduce $\frac{4}{n}G^{(\tau)}$ and apply [Algorithm 1 \(Theorem 5.1\)](#): the p output basis vectors of minimal norm, pulled back to $M_2(R_n)$ and taken with both signs, are the $2(n/2 + 1)$ shortest vectors of $\Lambda_B^{(\tau)}$. This uses $\text{poly}(n) \times \text{SVP}(n/2 + 1)$.
3. For each of the two Y with $Y \equiv I \pmod{2}$ (exactly $\pm V_\tau$, by the parity test of [Appendix D](#)): compute a $\mathbb{Z}$ -basis and T -Gram of $\mathcal{M}_Y$, $T(x, y) := \text{Tr}_{K_n}(x^* Q y)$, by Hermite normal form on the constraint $\tau(x) = Y^{-1}x$ (with $Y^{-1} = \text{adj } Y \in M_2(R_n)$ since $\det Y = 1$); LLL-reduce $\frac{1}{n}T$ -Gram and solve $\mathbb{Z}^n$ -LIP on it via [\[Duc23\]](#), using $\text{poly}(n) \times \text{SVP}(n/2 + 1)$ ([Appendix D](#)). For each pair (u_1, u_2) from the resulting shortest-vector set with $u_1^* Q u_2 = 0$ (at most $\binom{n}{2}$ pairs up to sign), test whether $B' := [u_1 \ u_2]^{-1}$ satisfies $B' \in \text{GL}_2(R_n)$ and $B'^* B' = Q$; output the first such B' .

By [Proposition D.2](#), step (3) succeeds for $Y = \pm V_\tau$. □

6.1 Cost estimates

[Table 1](#) compares the per-call cost of the SVP oracle at the designers' blocksize β_{key} and at the oracle dimension $n/2 + 1$ of [Theorem 6.1](#), under both the Core-SVP heuristic ($2^{0.292\beta}$ [\[BDGL16\]](#)) and the gate-count model of [Albrecht, Gheorghiu, Postlethwaite, and Schanck](#) [\[AGPS20\]](#). Asymptotically the improvement is a factor of two in the oracle dimension: $\beta_{\text{key}}/n \rightarrow 1$ under the model of [\[HAWK25, §5.1\]](#) while ours is $n/2 + 1$. With the provable sieve of [\[ADRS15\]](#) as the oracle this gives time $2^{(n/2+1)+o(n)}$ unconditionally.

Table 1. Cost of one SVP-oracle call. β is the oracle dimension: β_{key} [\[HAWK25, Table 7\]](#) for the spec, $n/2 + 1$ for the attack. *Core-SVP is our conversion $2^{0.292\beta}$ (exponent rounded to the nearest integer); the specification reports blocksizes and AGPS20 gate counts only. AGPS20 gates are the classical list-decoding gate counts of [\[AGPS20\]](#) at sieve dimension $\beta_{\text{eff}} = \beta - \text{d4f}(\beta)$, with dimensions-for-free [\[Duc18\]](#) computed and rounded as in [\[HAWK25, §5.3\]](#). [†]HAWK-256 is a challenge parameter set [\[HAWK25, §3.2\]](#), for which the specification states no gate count (the entry 2^{74} at β_{key} is our computation); [§6.2](#) reports actual recoveries.

	n	β		Core-SVP*		AGPS20 gates	
		Spec	Attack	Spec	Attack	Spec	Attack
HAWK-256 [†]	256	211	129	2^{62}	2^{38}	2^{74}	2^{52}
HAWK-512	512	452	257	2^{132}	2^{75}	2^{141}	2^{86}
HAWK-1024	1024	940	513	2^{274}	2^{150}	2^{278}	2^{158}

The HAWK specification reports total key-recovery costs of 2^{150} and 2^{288} gates for HAWK-512 and HAWK-1024 [\[HAWK25, §5.3\]](#), computed as $n \cdot C_{\text{gate}}(\beta_{\text{eff}})$:

C_{gate} is the AGPS20 gate count of one sieve call [AGPS20], $\beta_{\text{eff}} = \beta_{\text{key}} - d4f(\beta_{\text{key}})$, and n is the spec’s lower bound on the number of oracle calls in one BKZ tour. Under the same model at our dimension $n/2 + 1$ (so $\beta_{\text{eff}} = 224$ and 464 after dimensions-for-free), the per-call costs are $2^{86.0}$ and $2^{157.8}$; multiplying by the call count below gives totals of at most 2^{108} and 2^{182} .

The reduction makes $O(n^2 \log n)$ oracle calls in at most three reduction runs: one Algorithm 1 run in step (2) and at most two $\mathbb{Z}^n$ -LIP runs in step (3), each preceded by LLL. After LLL every basis vector has squared norm at most 2^n , since $\|b_i\|^2 \leq 2^{D-1} \lambda_i(\Lambda)^2$ for rank D [LLL82, p. 518] and the successive minima λ_i are key-independent (at most $\sqrt{2}$ and 1 for steps (2) and (3) respectively, by Propositions 4.5 and D.2). At this input size, the iteration bound of Theorem 5.1 is $(n/2+1)^2 \ln(n/2+1) + O(n \log n)$ for step (2), and that of Remark D.3 is $(n+1)(n/2+1) \ln(n/2+1) + O(n \log n)$ per step-(3) run; each iteration makes two oracle calls, and the terminal HKZ reductions add $O(n)$. The totals come to at most $2^{21.9}$ and $2^{24.0}$ calls for HAWK-512 and HAWK-1024, with the input basis entering only through a doubly-logarithmic term contributing under 1%.

Only the SVP-oracle calls are priced, on both sides. The polynomial-time arithmetic outside the oracle (HNF, LLL, Gram updates, the parity tests of step (3)) is omitted, as is the non-oracle cost of a BKZ tour in the specification’s count. Neither side prices memory access, the progressivity factor of roughly 2^5 [SAB22, §5.2.1], or the overheads of [Duc22]; the latter two are explicitly set aside in [HAWK25, §5.3].

A heuristic estimate in the specification’s own methodology goes further: under the GSA-intersect model of Appendix C, progressive BKZ reaches the minimum of the rescaled cocycle lattice — a τ -cocycle, by Lemma 4.4 — at blocksize 205 (HAWK-512) and 432 (HAWK-1024) rather than the proven oracle dimension $n/2 + 1$, giving heuristic key-recovery costs of about $2^{80.8}$ and $2^{146.5}$ gates against the specification’s 2^{150} and 2^{288} . The provable totals above are thus conservative by roughly a further 2^{27} and 2^{35} ; Appendix C gives the conversion and its caveats.

6.2 Implementation and key recovery for HAWK-256

We implemented the attack as a released end-to-end program [code] and recovered the secret keys of two HAWK-256 public keys generated by the reference key generator [HAWK25]. The implementation follows steps (1)–(3) of Theorem 6.1, replacing the provable LIP solvers with heuristic lattice reduction and taking two shortcuts in step (3); the theorem’s cost bound does not depend on either.

Step (1) constructs the 256×256 integer Gram matrix of $\Lambda_B^{(\tau)}$ via an integer Hermite-normal-form kernel computation followed by LLL reduction of the resulting basis.

Step (2) BKZ-reduces this Gram matrix with fpylll to blocksize 44 and runs a progressive BGJ-type sieve [BGJ15], using the AMX implementation of Zhao, Ding, and Yang [ZDY25], to a maximum sieving dimension of 118. The sieve

returns two shortest vectors Y, Y' of $\Lambda_B^{(\tau)}$ not related by sign. A sieve database contains some shortest vectors long before it provably contains all of them; Remark 6.2 is what makes two enough.

Remark 6.2 (intertwiner sublattice). Let Y, Y' be shortest vectors of $\Lambda_B^{(\tau)}$ with $Y \neq \pm Y'$ and neither equal to $\pm V_\tau$, and set $M := Y \cdot \text{adj}(Y')$. The rank- $n/2$ sublattice

$$\Lambda_a := \{X \in \Lambda_B^{(\tau)} : MX = X\tau(M)\}$$

is cut out by one further public linear system (its coefficients involve only Q, Y, Y'), and its two shortest vectors are $\pm V_\tau$ (proof in Appendix D). The hypothesis that neither sieve output equals $\pm V_\tau$ is testable by the parity test of Appendix D; when it fails the construction is unnecessary.

The implementation locates $\pm V_\tau$ directly from Λ_a : it computes an explicit $\mathbb{Z}$ -basis of this rank-128 lattice and runs LLL followed by progressive BKZ; blocksize at most 24 sufficed in every run. It then reaches an equivalent key by a τ -tower descent from $\ell = 9$ down to $\ell = 3$ (Appendix D, after Lemma D.4) in place of the single $\mathbb{Z}^n$ -LIP solve of step (3); each level re-runs steps (1)–(2) at half the degree, and the resulting rank-128 cocycle lattice at the $\ell = 8$ level was solved at BKZ blocksize at most 16, every lower level by LLL alone. We make no provable claim for the tower descent; the provable path remains the single $\mathbb{Z}^n$ -LIP solve.

Every attempt succeeded. For one of the two keys, two independent runs each produced a solution: distinct but unitarily-equivalent secret keys, both passing the reference implementation’s sign/verify check. A recovery completes within a few hours on a single 96-core Sapphire Rapids server, dominated by the step-(2) sieve. The release is a single-command pipeline with per-stage checkpoints that verifies its output against the HAWK reference implementation as its final step; Appendix A lists the lemma checks and the (heuristic) end-to-end runs at small ℓ . For HAWK-512 the proven oracle dimension is 257; we have not attempted it.

7 Scope of the Attack

The attack requires two ingredients: a τ -cocycle lattice (§4), and a half-dimension reduction for its isometry class (§5, Appendix E). The first is settled by:

Theorem 7.1. *An order-2 Galois element $\tau \neq c$ of $\mathbb{Q}(\zeta_m)$ — and hence the construction of §4 — exists iff $(\mathbb{Z}/m)^\times$ is non-cyclic, iff $m \notin \{1, 2, 4, p^k, 2p^k\}$ for p an odd prime.*

Proof. §4 requires an order-2 Galois element $\tau \neq c$; by Lemma B.1 one exists iff $(\mathbb{Z}/m)^\times$ is non-cyclic. $\square$

For the second, Theorem 6.1 is proven only for powers of two, but for every non-cyclic m the cocycle lattice has V_τ as a shortest vector with minimum $\varphi(m)/4$ (the proof of Lemma 4.4 uses only $\tau^2 = 1, \tau \neq c$), and we expect heuristic lattice

reduction to find it at blocksize somewhat below $\varphi(m)/2$ (roughly 0.36–0.42 $\varphi(m)$ at the power-of-two parameter sets under the model of Appendix C) whenever m has a small prime factor: preliminary computations at small parameters for $m = 4p$ and $m = 3 \cdot 2^\ell$ suggest the cocycle lattice satisfies the hypothesis of [BN24, Thm 1] with the same constant $\lambda_1 \lambda_1^\vee \rightarrow 1/\sqrt{2}$ as the power-of-two case.

Conductors $m \in \{p^k, 2p^k\}$ (and the degenerate $m \leq 4$) evade the construction. Over those fields the rank- $3\varphi(m)/2$ c -cocycle of §8.3 still applies, with blocksize roughly $0.72 \beta_{\text{key}}$ under the same model (Appendix C).

8 Relation to Prior Work

8.1 Subfield attacks

The NTRU subfield attack of Albrecht, Bai, and Ducas [ABD16] norms the public key $h \mapsto N_{K_n/E}(h)$ to a subfield E ; the subring variant of Kirchner and Fouque [KF17] instead restricts the multiplication-by- h matrix to a subring of the same index. Kirchner and Fouque prove a complexity bound [KF17, Thm 7] that depends on the secret width only through $n \log \sigma_{\text{sk}}$ (where σ_{sk} is the width of the NTRU secret), a quantity they observe is invariant under subfield descent [KF17, §4.2], so descent gains nothing at non-overstretched parameters. By contrast, the cocycle V_τ of §4 has $\det V_\tau = 1$ and $Q^{(\tau)}(V_\tau) = n/4$ regardless of the key-generation distribution (Lemma 4.4), so the conservation argument has no analogue; the construction of §4 involves neither a modulus nor a subfield descent.

8.2 Gentry–Szydlo and quaternion methods

Gentry and Szydlo [GS02, §8] propose extending their algorithm to recover f from a basis of (f) and the relative norm of f over an index-2 subfield only “where the degree-2 extension is complex conjugation”. Every subsequent module-LIP reduction we are aware of — to a relative- or reduced-norm principal-ideal problem [MPPW24; Che25; APvW25], via the symplectic automorphism [LJPW24], or via the Hermitian square-basis oracle of [EP24] — likewise uses no Galois involution other than complex conjugation. The construction of this paper instead uses the involution τ , whose fixed field $K_n^\tau = \mathbb{Q}(\zeta_{2^\ell-1})$ is complex cyclotomic; the resulting public lattice has rank n rather than $2n$, and the reduction target is SVP in dimension $n/2 + 1$ rather than a principal-ideal problem.

8.3 Comparison with the c-cocycle

Chevignard et al. [Che25, §1] construct a rank- $2n$ lattice from the public key whose shortest vector is the secret (their d is our n); for HAWK it is hypercubic, and they show a single SVP call in dimension $2n$ breaks the scheme. Replacing

τ by c throughout §4 gives a related c -cocycle lattice of rank $3n/2$: constraint (C2) degenerates to “ QY symmetric” (since $\sigma = c^2 = \text{id}$), so it is not the lattice of [Che25], which imposes neither (C1) nor (C2). For $\tau \neq c$, $\sigma = c\tau \neq \text{id}$ is itself an order-2 automorphism and (C2) is an independent $n/2$ -dimensional restriction; this rank saving is the source of the factor-2 blocksize improvement. The third involution σ gives the same rank n , since the proofs use only $\tau^2 = 1$, $c\tau \neq 1$, $[K_n : K_n^\tau] = 2$, and the power-of-two trace orthogonality, all of which hold with $\tau \leftrightarrow \sigma$. Higher-order Galois elements admit no adj-linearized analogue of (C1), since the linearization of $\gamma(Y) = Y^{-1}$ requires $\gamma^2 = \text{id}$; we found no useful replacement.

8.4 The reductions of Ducas and of Bambury–Nguyen

Ducas [Duc23] solves $\mathbb{Z}^d$ -LIP with polynomially many SVP-oracle calls in dimension $\lfloor d/2 \rfloor + 1$, by a two-block reduction whose progress lemma [Duc23, Lemma 4] bounds the per-round drop of an integer potential. Bambury and Nguyen [BN24] extend the early-abort form of the reduction to any sublattice of $\mathbb{Z}^D$ with $\lambda_1(A)\lambda_1(A^\vee) < 1 - \varepsilon$, returning one short primal or dual vector at the same oracle dimension. As noted in §5, we run Ducas’s algorithm on the class $[A']$, where his progress lemma fails and Appendix E supplies the replacement; Remark E.5 compares the two routes in detail.

8.5 Lattice automorphisms and van Gent–Pulles

Van Gent and Pulles [GP25] show that, given any $\mathbb{Z}$ -automorphism of the public form outside the trivial group generated by the roots of unity and the symplectic automorphism of [LJPW24], smLIP reduces to BKZ on a kernel sublattice of half the rank: their Propositions 1–2 construct the kernel and recover the key from a short vector of it, deterministically and in polynomial time, and their Theorem 1 estimates the BKZ cost heuristically. §5 and Appendix D run their descent on the automorphisms $x \mapsto Y\tau(x)$ that the shortest vectors of the cocycle lattice supply (Lemma D.1); Appendix D identifies the kernel exactly as $\sqrt{n}\mathbb{Z}^n$, which lets Ducas’s algorithm replace the heuristic BKZ step.

9 Discussion and Open Questions

One attempt to halve the oracle dimension again, to $n/4 + 1$, by running Algorithm 1 at cut $\lfloor p/2 \rfloor$, is blocked: the loop reaches the non-terminal fixed point $S = \mathbb{Z}^{\lfloor p/2 \rfloor} \subset \mathbb{Z}^p$ (where $\text{vol}(S) = 1$ and the product-progress quantity of Lemma E.2 equals 1), and for $n > 8$ the residual back-block LIP still requires $\text{SVP}(n/2 - 1)$ by duality and Corollary E.4. As a generic $[A']$ -LIP problem this would require a gap- $\sqrt{2}$ Rankin solver below half-dimension, for which no technique is known.

The HAWK instance carries structure a generic member of $[A']$ lacks: $A_B^{(\tau)}$ is publicly a rank-4 quadratic $\mathcal{O}_F$ -lattice (for $f \in \mathcal{O}_F$, both (C1) and (C2) commute

with multiplication by f , and $\varphi_\tau(fY) = f\varphi_\tau(Y)$), with $\mathcal{O}_F$ -valued form $\det Y$. Since F is totally real its only roots of unity are ± 1 , which appears to rule out the known structured-sieve speedups; we leave open whether this rank-4 module structure admits a solver with SVP-oracle dimension below $n/2$.

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A Computational Verification

The released code [code] is the heuristic implementation of §6.2, not the provable reduction of §5: its SVP stage is progressive BKZ plus a sieve. An accompanying script re-derives the §4 statements exactly on a random key at $\ell \leq 6$: Lemmas 4.1–4.4 and Proposition 4.5, including the isometry class $\sqrt{n/4} \mathbb{Z}^{n/2+1} \oplus \sqrt{n/2} \mathbb{Z}^{n/2-1}$ and the shortest-vector count $2(n/2 + 1)$ by exact enumeration. End-to-end, the same code recovers keys on synthetic instances at $\ell = 5, 6, 7, 8$ in 0.7 s, 1.4 s, 19 s, and 8.6 minutes respectively (single core; BKZ alone reaches the minimum for $n \leq 128$), and on HAWK-256 keys at $\ell = 9$. Instances at $\ell \leq 8$ are synthetic (a random $\text{SL}_2(R_n)$ basis from a reimplemented generator); at $\ell = 9$ keys are generated, and every recovered key is checked by a sign/verify round-trip, using the official HAWK submission package [HAWK25].

B Involutions of $(\mathbb{Z}/m)^\times$

The following lemma is the group-theoretic input to Theorem 7.1; we include its proof for completeness.

Lemma B.1. *For $m \geq 1$ the following are equivalent: (i) $(\mathbb{Z}/m)^\times$ contains an element of order 2 other than -1 ; (ii) $(\mathbb{Z}/m)^\times$ is non-cyclic; (iii) $m \notin \{1, 2, 4, p^k, 2p^k\}$ for p an odd prime.*

Proof. By the Chinese Remainder Theorem, $(\mathbb{Z}/m)^\times \cong \prod_i (\mathbb{Z}/p_i^{k_i})^\times$ over the prime-power factors $p_i^{k_i}$ of m . Each factor with p_i odd is cyclic (the primitive root theorem) of order $p_i^{k_i-1}(p_i - 1)$, which is even; $(\mathbb{Z}/2)^\times$ is trivial, $(\mathbb{Z}/4)^\times$ is cyclic of order 2, and $(\mathbb{Z}/2^k)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/2^{k-2}$ is non-cyclic for $k \geq 3$.

(iii) $\Rightarrow$ (i): if $m \notin \{1, 2, 4, p^k, 2p^k\}$ then either $8 \mid m$, in which case the factor $(\mathbb{Z}/2^k)^\times$ with $k \geq 3$ already contains $(\mathbb{Z}/2)^2$ in its 2-torsion, or the decomposition has at least two factors of even order (this happens when two distinct odd primes divide m , or when one odd prime divides m and $4 \mid m$). When there are at least two even-order factors, the element that reduces to -1 in one of them and to $+1$ in every other factor has order 2 and differs from -1 in a second even-order factor. Either way the 2-torsion subgroup has order at least 4 and contains an involution other than -1 .

(i) $\Rightarrow$ (ii): for $m > 2$ the class of -1 has order 2, so (i) provides two distinct elements of order 2; a cyclic group has at most one. (For $m \leq 2$ the group is trivial and (i) is false.)

(ii) $\Rightarrow$ (iii): contrapositively, for $m \in \{1, 2, 4, p^k, 2p^k\}$ the decomposition has at most one nontrivial factor, which is cyclic. $\square$

C The $0.72\beta_{\text{key}}$ estimate of §7

Throughout this appendix m is the conductor and $d_+ := [K_n^+ : \mathbb{Q}] = \varphi(m)/2$ ($d_+ = n/2$ for the power-of-two parameter sets). The blocksize model is the

heuristic GSA-intersect estimate (GSA: Geometric Series Assumption), as used for [HAWK25, Table 8]: an unusually short vector of length ν in a rank- d lattice of volume $\mathcal{V}$ is found by progressive BKZ at the smallest β with

$$\sqrt{\beta/d} \nu \leq \delta_\beta^{2\beta-d-1} \mathcal{V}^{1/d},$$

$$\delta_\beta = \left(\frac{\beta}{2\pi e} \right)^{1/(2(\beta-1))},$$

a condition that depends only on d and the gap $\ln \nu - \frac{1}{d} \ln \mathcal{V}$; the gap-0 instance is [DPPvW22, eq. (4)].

For every conductor, the c -cocycle parametrization at $B = I$ is $W = \begin{pmatrix} a & b \\ b & c(a) \end{pmatrix}$ with $a \in R_n$, $b \in R_n^{c-1}$, $\det W = |a|^2 + |b|^2$, and $Q^{(c)}(W) = \text{Tr}_{K_n^+}(\det W) = \frac{1}{2} \langle a, a \rangle_{K_n} + \frac{1}{2} \langle b, b \rangle_{K_n}$ (using $|a|^2, |b|^2 \in K_n^+$ and $[K_n : K_n^+] = 2$). As in Lemma 4.4 with K_n^+ in place of F , the minimum is $\text{Tr}_{K_n^+}(1) = d_+$, attained at the secret cocycle $B^{-1}c(B)$, so $\nu = \sqrt{d_+}$.

Power-of-two conductors. The key lattice is a scaled $\mathbb{Z}^{2n}$, gap 0; the model gives $\beta_{\text{key}} = 221/457/933$, within ± 10 of the simulated values $211/452/940$ of [HAWK25, Table 7] and within ± 1 of [HAWK25, Table 8]. For the c -cocycle lattice, the trace-orthogonal power basis (§3.3) gives $(R_n, \langle \cdot, \cdot \rangle_{K_n}) = \sqrt{n} \mathbb{Z}^n$, and c acts on that basis as a signed permutation with one fixed point (1), one antifixed point ($\zeta^{n/2}$), and $d_+ - 1$ sign-2-cycles, giving $(R_n^{c=-1}, \langle \cdot, \cdot \rangle_{K_n}) = \sqrt{n} \mathbb{Z} \oplus \sqrt{2n} \mathbb{Z}^{d_+-1}$ by the per-cycle accounting of the proof of Proposition 4.5. Hence $\Lambda_I^{(c)} \cong \sqrt{d_+} \mathbb{Z}^{2d_++1} \oplus \sqrt{2d_+} \mathbb{Z}^{d_+-1}$, $\text{gap} \approx -\frac{1}{6} \ln 2$.

Power-of-two τ -cocycle. The same model applies to the rescaled cocycle lattice of Corollary 4.6, $\Lambda' = \mathbb{Z}^{n/2+1} \oplus \sqrt{2} \mathbb{Z}^{n/2-1}$: rank $d = n$, $\nu = 1$, $\mathcal{V} = 2^{(n/2-1)/2}$, so the gap is $-\frac{n/2-1}{2n} \ln 2 \approx -\frac{1}{4} \ln 2$. The model then gives block sizes $92/205/432$ at HAWK-256/-512/-1024, that is 0.42–0.46 of the same-model β_{key} above and somewhat below the oracle dimension $n/2 + 1$ of Theorem 6.1; the next paragraph converts these into the gate counts quoted in §6.1. The ± 10 calibration against the simulated block sizes is for the key lattice only; a simulator run on the class $[\Lambda']$, as was done for the key lattice in [HAWK25, Table 7], would be the proper counterpart.

Odd-prime-power conductors. The power basis is not orthogonal, so neither lattice has gap 0. The key-lattice minimum is $\varphi(m)$: for $x \in R_n \setminus \{0\}$, $\text{Tr}_{K_n}|x|^2 = \sum_\sigma |\sigma(x)|^2 \geq \varphi(m) |N(x)|^{2/\varphi(m)} \geq \varphi(m)$ by AM–GM and $|N(x)| \geq 1$, with equality iff x is a root of unity. The volumes follow from $|\text{disc } K_n|$ and a closed form for $\det(R_n^{c=-1}, \langle \cdot, \cdot \rangle_{K_n})$, both checked against exact linear algebra at $m \in \{5, 7, 9, 11, 13, 25, 27\}$.

Ratio. Across every conductor we computed, $\beta_{c\text{-cocycle}}/\beta_{\text{key}} \in [0.69, 0.73]$: 0.71–0.72 at HAWK-512 and HAWK-1024, 0.72–0.73 for odd-prime-power conductors of comparable degree ($m = 769, 1153, 3^7, 5^5, 2309$), and 0.69–0.73 at HAWK-256 depending on whether the denominator is the simulated or same-model β_{key} . The ratio $3/4$ suggested by the ranks alone is corrected downward

chiefly by the $2^{1/6}$ volume excess: the cocycle target is relatively shorter in its lattice than the unit vectors are in the key lattice.

Gate-count conversion for §6.1. At the power-of-two τ -cocycle block sizes above (205 for HAWK-512, 432 for HAWK-1024), the recipe of [HAWK25, §5.3] converts as follows: dimensions-for-free $\text{d4f}(\beta) = \beta \ln(4/3) / \ln(\beta/(2\pi e))$, rounded, with β_{eff} lowered to a multiple of 8, gives 176 and 392; the classical list-decoding gate counts of [AGPS20] are $2^{71.8}$ and $2^{136.5}$ per call; and multiplying by n for one tour gives the heuristic totals $2^{80.8}$ and $2^{146.5}$ quoted in §6.1. Isolating $\pm V_\tau$ from the shortest vectors (Remark 6.2) is cheaper by the same recipe. The estimate counts only the top-level step-(2) reduction and prices the final descent by the τ -tower of §6.2; the alternative, a direct reduction on the $\mathbb{Z}^n$ -LIP kernel of step (3), would cost more ($2^{85.5}$ and $2^{153.6}$). Each tower level repeats the top-level computation at half the degree and is cheaper by the same recipe (the $n = 256$ level costs about 2^{51} , the $n = 512$ level about 2^{81}), so the top level dominates the tower. The block sizes are GSA-intersect estimates rather than the simulated ones behind [HAWK25, Table 7], a difference of about ± 10 in β , or ± 2.4 bits; memory, the progressivity factor, and the overheads of [Duc22] remain unpriced on both sides, as in §6.1.

D The Descent as a Strengthened Analysis of van Gent–Pulles

Throughout, Y denotes one of the $2(n/2 + 1)$ shortest vectors of $\Lambda_B^{(\tau)}$, so that $Y\tau(Y) = I$ and $\det Y = 1$ (Lemma 4.4(ii)), hence $Y^{-1} = \tau(Y)$, and $Y^*QY = \tau(Q)$: by (C2), $Y^*QY = Y^*\sigma(Y)^T\tau(Q)$, and $Y^*\sigma(Y)^T = (\sigma(Y)c(Y))^T = \sigma(Y\tau(Y))^T = I$ using $c\tau = \sigma$ and (C1).

Lemma D.1. *The map $U_Y : R_n^2 \rightarrow R_n^2$, $x \mapsto Y\tau(x)$, is an order-2 automorphism of the quadratic $\mathbb{Z}$ -module (R_n^2, T) , $T(x, y) = \text{Tr}_{K_n}(x^*Qy)$, lying outside the trivial group G_n of [GP25] (generated by the roots of unity and the symplectic automorphism, the latter included after [LJPW24]); and $\ker(U_Y - 1) = \mathcal{M}_Y$, $\ker(U_Y + 1) = \mathcal{M}_{-Y}$.*

Proof. Order 2: $U_Y^2(x) = Y\tau(Y)x = x$ by the cocycle equation. Isometry: $T(U_Yx, U_Yy) = \text{Tr}_{K_n}(\tau(x)^*Y^*QY\tau(y)) = \text{Tr}_{K_n}(\tau(x)^*\tau(Q)\tau(y)) = \text{Tr}_{K_n}(\tau(x^*Qy)) = T(x, y)$, using $Y^*QY = \tau(Q)$, $\tau(x)^* = \tau(x^*)$ (the involutions c and τ commute), and the Galois-invariance of the trace. Exclusion: U_Y is τ -semilinear ($U_Y(\alpha x) = \tau(\alpha)U_Y(x)$ for $\alpha \in R_n$), while every element of G_n is R_n -linear and $\tau \neq \text{id}$. Kernels: $U_Yx = \pm x \iff \tau(x) = (\pm Y)^{-1}x \iff x \in \mathcal{M}_{\pm Y}$, using $Y^{-1} = \tau(Y)$. $\square$

In the terminology of [GP25], U_Y lies in the orthogonal group of the public form outside their trivial group, $\mathcal{M}_{\pm Y} = \ker(U_Y \mp 1)$ are the kernel sublattices of [GP25, Prop. 2], and [GP25, Prop. 1] — proved there, deterministic polynomial time — recovers an equivalent key from any sufficiently short kernel vector. [GP25, Lem. 3 and Prop. 2] prove that these kernels are of the form $\mathbb{Z}^s \oplus \sqrt{2}\mathbb{Z}^t$

for some $s, t \geq 0$ with $\lambda_1 \leq \sqrt{2}$ — enough for the heuristic pricing of [GP25, Thm. 1], but not enough to determine the split (s, t) , which differs between $Y = \pm V_\tau$ and the other candidates. The proofs below identify the kernels exactly — $(\mathcal{M}_{\pm V_\tau}, T) \cong \sqrt{n} \mathbb{Z}^n$, with the columns of B^{-1} up to roots of unity as shortest vectors — and this exactness is what admits the provable solver of [Duc23] (Remark D.3) and so upgrades the endgame from the heuristic pricing of [GP25, Thm. 1] to the unconditional accounting of Theorem 6.1.

Proposition D.2. *Let $Y \in \Lambda_B^{(\tau)}$ be a shortest vector with $\varphi_\tau(Y) = \pm I$. Then:*

1. $\mathcal{M}_Y = B^{-1} \mathcal{O}_{K_n^\tau}^2$ (resp. $B^{-1}(\zeta \mathcal{O}_{K_n^\tau})^2$ for $-I$), a $\mathbb{Z}$ -lattice of rank n ;
2. $(\mathcal{M}_Y, T) \cong \sqrt{n} \mathbb{Z}^n$;
3. the shortest vectors of $(\mathcal{M}_Y, T)$ are $\{\pm B^{-1} \zeta^{2j} e_k : 0 \leq j < n/2, k \in \{1, 2\}\}$ (resp. $\{\pm B^{-1} \zeta^{2j+1} e_k\}$ in the $-I$ case), where e_1, e_2 is the standard basis of R_n^2 ; any pair (u_1, u_2) from this set with $u_1^* Q u_2 = 0$ satisfies $[u_1 \ u_2]^{-1} \in U_2(R_n) \cdot B$, an equivalent HAWK secret key. (Here $[u_1 \ u_2]$ is the matrix with columns u_1, u_2 .)

Proof. (1) For $Y = V_\tau$: $\tau(x) = V_\tau^{-1}x = \tau(B)^{-1}Bx \Leftrightarrow \tau(Bx) = Bx \Leftrightarrow Bx \in (K_n^\tau)^2$. Intersecting with R_n^2 and using $BR_n^2 = R_n^2$ ($B \in \text{SL}_2(R_n)$) and $R_n \cap K_n^\tau = \mathcal{O}_{K_n^\tau}$: $\mathcal{M}_Y = B^{-1} \mathcal{O}_{K_n^\tau}^2$. For $Y = -V_\tau$, $\tau(Bx) = -Bx$ gives $\mathcal{M}_Y = B^{-1}(\zeta \mathcal{O}_{K_n^\tau})^2$. (2) For $x = B^{-1}y$: $x^* Q x = y^* (B^{-1})^* Q B^{-1} y = y^* y$, so $T(x, x') = \text{Tr}_{K_n}(y^* y')$. On the basis $\{\zeta^{2j} e_k : 0 \leq j < n/2, k \in \{1, 2\}\}$ of $\mathcal{O}_{K_n^\tau}^2$: $\text{Tr}_{K_n}(\zeta^{2j} \zeta^{2j'}) = n \delta_{j,j'}$. Hence Gram $= nI_n$. (3) The shortest vectors of a lattice in class $[\mathbb{Z}^m]$ are its orthogonal $\mathbb{Z}$ -basis up to sign, so here $\{\pm B^{-1} \zeta^{2j} e_k\}$. For any two of these, $(B^{-1} \zeta^{2j} e_k)^* Q (B^{-1} \zeta^{2j'} e_{k'}) = \zeta^{2(j'-j)} \delta_{k,k'}$, which vanishes iff $k \neq k'$. For such a pair, $[u_1 \ u_2] = B^{-1}M$ with M a monomial matrix whose nonzero entries are $2n$ -th roots of unity, hence $M \in U_2(R_n)$ and $[u_1 \ u_2]^{-1} \in U_2(R_n) \cdot B$. $\square$

Selecting $\pm V_\tau$: the parity test. The other n shortest vectors do not lead to the key: for Y with $\varphi_\tau(Y) = W(0, \omega)$, $\omega = \zeta^j$ with j odd (proof of Lemma 4.4(ii)), the substitution $y = Bx$ of Proposition D.2's proof turns the defining condition of $\mathcal{M}_Y$ into $y_2 = -\omega^{-1} \tau(y_1)$ with $y_1 \in R_n$ free, and $T(x, x') = 2 \text{Tr}_{K_n}(\overline{y_1} y'_1)$, so $(\mathcal{M}_Y, T) \cong \sqrt{2n} \mathbb{Z}^n$, whose shortest vectors carry no orthonormal pair. The two useful candidates are recognized by reduction modulo 2: since $\tau(\zeta^i) = (-1)^i \zeta^i \equiv \zeta^i$, the involution τ acts trivially on $R_n/2R_n$, so $\tau(B) \equiv B$ and $\pm V_\tau = \pm B^{-1} \tau(B) \equiv I \pmod{2}$; whereas any other shortest vector $Y = B^{-1}W(0, \omega) \tau(B) \equiv B^{-1}W(0, \omega) B \not\equiv I \pmod{2}$, because B is invertible modulo 2 (as $\det B = 1$), so conjugation by it is a bijection of $M_2(R_n/2R_n)$ fixing I , and $W(0, \omega)$ has zero diagonal. Thus exactly the two candidates $\pm V_\tau$ satisfy $Y \equiv I \pmod{2R_n}$, an $O(n)$ -coefficient check. (The reduction does require the exact shortest vectors: $\mathcal{M}_Y \neq 0$ iff $\det Y = 1$, so an approximate-SVP output Y' with $Q^{(\tau)}(Y') > n/4$ gives $\mathcal{M}_{Y'} = \{0\}$.)

Remark D.3 (improved iteration count for [Duc23]). The two-bound termination analysis of Theorem 5.1 (in the notation of its proof, §E.4: potential W_t , primal factor α) applies to [Duc23, Alg. 1] on a lattice in $[\mathbb{Z}^d]$,

$d = 2k' + 1$: after the primal step the front block $\mathcal{L}(B_{[0:k']})$ is a primitive rank- $(k'+1)$ sublattice of volume $W_t^{1/2}\alpha$ with $\alpha \leq 1$ [Duc23, Lemma 3], its dual has covolume $(W_t^{1/2}\alpha)^{-1}$, and Minkowski's theorem gives the same recursion $W_{t+1} \leq (k'+1)W_t^{1-1/(k'+1)}$; the basic per-iteration factor is $1 - 1/d$ [Duc23, Lemma 4]. The iteration count of [Duc23, Thm 2] therefore improves to $d(1 + (k'+1)\ln(k'+1)) + (k'+1)\ln(2k'\ln\max_i \|b_i\|) + 3(k'+1)$ for any input basis, again doubly logarithmic in the input. Applied through the even-rank adjustment $\Lambda \mapsto \Lambda \oplus \mathbb{Z}$ of [Duc23] (the rank- n lattice $(\mathcal{M}_Y, \frac{1}{n}T)$ becomes the odd rank $n+1$), it gives $(n+1)(n/2+1)\ln(n/2+1) + O(n\log n)$ iterations per descent run under the LLL input bound of §6.1.

Lemma D.4 (the τ -tower step). Let $Y \in \Lambda_B^{(\tau)}$ be a shortest vector with $\varphi_\tau(Y) = \pm I$ (so $Y\tau(Y) = I$, $\det Y = 1$, and $Y \equiv I \pmod{2}$), and write $K' := K_n^\tau$, $\mathcal{O}' := \mathcal{O}_{K'}$, so that $R_n = \mathcal{O}' \oplus \zeta\mathcal{O}'$. Then (i) $\mathcal{M}_Y$ is the $\mathcal{O}'$ -module generated by the columns of $(I+Y)/2$ and $(I-Y)/(2\zeta)$ (both in $M_2(R_n)$), and (ii) the alternating $\mathcal{O}'$ -bilinear form $\mu(x, y) := \frac{1}{2}\text{Tr}_{K_n/K'}(\det[x \ y])$ on $\mathcal{M}_Y$ is unimodular: $\mu(f_1, f_2) \in \mathcal{O}'^\times$ for every $\mathcal{O}'$ -basis (f_1, f_2) . Consequently any pair $(x, y) \in \mathcal{M}_Y^2$ with $\mu(x, y) = 1$ is a free $\mathcal{O}'$ -basis of $\mathcal{M}_Y$.

Proof. (i) The matrices $(I+Y)/2$ and $(I-Y)/(2\zeta)$ are integral because $Y \equiv I \pmod{2}$ and ζ is a unit. For $a \in \mathcal{O}^2$: $\tau(\frac{I+Y}{2}a) = \frac{I+Y^{-1}}{2}a = Y^{-1}\frac{I+Y}{2}a$, and likewise for $(I-Y)/(2\zeta)$ using $\tau(\zeta) = -\zeta$, so both columns lie in $\mathcal{M}_Y$. Conversely, for $x \in \mathcal{M}_Y$ write $x = a + \zeta b$ with $a, b \in \mathcal{O}^2$; averaging $x = a + \zeta b$ with $x = Y\tau(x) = Ya - \zeta Yb$ gives $x = \frac{I+Y}{2}a + \frac{I-Y}{2\zeta}(\zeta^2 b)$ with $\zeta^2 b \in \mathcal{O}^2$. (ii) By Proposition D.2(1), $\mathcal{M}_Y = B^{-1}\Omega\mathcal{O}^2$ with $\Omega = 1$ (resp. $\Omega = \zeta$ for $-I$). For $x = B^{-1}\Omega u$, $y = B^{-1}\Omega w$ with $u, w \in \mathcal{O}^2$, since $\det B = 1$ we get $\det[x \ y] = \Omega^2 \det[u \ w] \in K'$, hence $\mu(x, y) = \Omega^2 \det[u \ w]$ ($\det[x \ y]$ is already τ -fixed). For an $\mathcal{O}'$ -basis $(f_1, f_2) = (B^{-1}\Omega u_1, B^{-1}\Omega u_2)$ we have $[u_1 \ u_2] \in \text{GL}_2(\mathcal{O}')$, so $\mu(f_1, f_2) = \Omega^2 \det[u_1 \ u_2] \in \mathcal{O}'^\times$. Finally, $\mu(x, y) = \det(C)\mu(f_1, f_2)$ where $C \in M_2(\mathcal{O}')$ expresses (x, y) in the basis; if $\mu(x, y) = 1$ then $\det C \in \mathcal{O}'^\times$ and (x, y) is a basis. $\square$

For the other shortest vectors (the case $\varphi_\tau(Y) = W(0, \omega)$, $\omega = \zeta^j$, above) the same computation gives μ with values in $2\zeta^{1-j}\mathcal{O}'$, never a unit — a second reason that only $\pm V_\tau$ descend.

The τ -tower descent (§6.2). The implementation replaces the single $\mathbb{Z}^n$ -LIP solve of step (3) by a descent through the tower of τ -fixed subfields $\mathbb{Q}(\zeta) \supset \mathbb{Q}(\zeta^2) \supset \mathbb{Q}(\zeta^4) \supset \dots$, i.e. from the conductor exponent ℓ down to 3. Each level computes an $\mathcal{O}_{K_n^\tau}$ -basis of $\mathcal{M}_{V_\tau} = B^{-1}\mathcal{O}_{K_n^\tau}^2$: by (i), the module is generated by the columns of $(I+V_\tau)/2$ and $(I-V_\tau)/(2\zeta)$, and by (ii), a free basis is any pair (x, y) with $\mu(x, y) = 1$, obtained by a Bézout step in $\mathcal{O}'$ (no class-group computation). The level then Gauss-reduces this basis under the Q -Hermitian form and recurses on the resulting HAWK public key over K_n^τ , re-running steps (1)–(2) there to obtain the next level's V_τ .

Proof of Remark 6.2. Recall $M := Y \text{adj}(Y')$ and $\Lambda_a := \{X \in \Lambda_B^{(\tau)} : MX = X\tau(M)\}$ for shortest vectors $Y \neq \pm Y'$ of $\Lambda_B^{(\tau)}$, neither equal to $\pm V_\tau$. Transporting

by the isometry φ_τ of Lemma 4.2, the condition $MX = X\tau(M)$ becomes $NW = W\tau(N)$ with $N = \varphi_\tau(Y) \cdot \text{adj}(\varphi_\tau(Y'))$; since $\varphi_\tau(Y), \varphi_\tau(Y')$ are shortest vectors other than $\pm I$, they have the form $W(0, b)$ with $b = \zeta^k$, k odd (§4.3; proof of Lemma 4.4(ii)), so N is diagonal. For $W = W(a, b)$ the diagonal entries of $NW - W\tau(N)$ vanish identically and the off-diagonal entries vanish only for $b = 0$ (using $Y \neq \pm Y'$), so $\varphi_\tau(\Lambda_a) = \{W(a, 0) : a \in R_n^\sigma\}$, the summand of $\Lambda_I^{(\tau)}$ containing $\pm I$, whose two shortest vectors are $\pm I$ by Proposition 4.5. $\square$

E Correctness of Block Reduction on $[\Lambda']$

This appendix proves that LIP for the class $[\Lambda'] = [\mathbb{Z}^p \oplus \sqrt{2}\mathbb{Z}^q]$ is solvable with polynomially many calls to exact SVP in dimension p (Theorem 5.1). The algorithm is Lucas’s two-block reduction [Duc23, Alg. 1], run unchanged on $[\Lambda']$ — the same primal/dual loop at the same oracle dimension, with only the cut fixed at $k := p - 1$ for this even-rank class and a final size-reduction made explicit; what is new is the analysis, since Lucas’s integrality and progress lemmas are proved only for $\mathbb{Z}^d$ (Lemmas E.1–E.2 replace them, and this appendix re-derives termination and correctness). The missing ingredient is the no-abort product-progress bound for $[\Lambda']$. In outline: the algorithm maintains a basis of the lattice and alternates two SVP-oracle steps on projected blocks of dimension at most p : a *primal* step that makes the Gram–Schmidt length $\|b_{p-1}^*\|$ as short as possible and a *dual* step that makes it as long as possible. The squared volume of the first $p - 1$ basis vectors is a positive integer (Lemma E.1) that each primal-plus-dual round multiplies by a factor at most $1 - 1/p$ (Lemma E.2), so after $O(n^2 \log n)$ rounds (for polynomially-bounded input) it reaches 1, at which point the front block spans a copy of $\mathbb{Z}^{p-1}$ and the hidden orthogonal basis can be read off (Theorem 5.1).

Notation. A sublattice S of a lattice Λ is *primitive* if $S = \text{span}(S) \cap \Lambda$; $\text{vol}(S)$ is its covolume (root-determinant of its Gram), π_U (for a subspace U) is orthogonal projection onto U , $\pi_S^\perp$ is orthogonal projection onto $\text{span}(S)^\perp$, and for S primitive in Λ one has $S^\vee = \pi_{\text{span}(S)}(\Lambda^\vee)$. For a basis $B = (b_0, \dots, b_{d-1})$ (unrelated to the secret key of §3.1) of a lattice $\mathcal{L}(B)$, b_i^* denotes the i -th Gram–Schmidt vector (the star is unrelated to the adjoint $*$) and $B_{[i:j]} := (\pi_{\{b_0, \dots, b_{i-1}\}}^\perp(b_i), \dots, \pi_{\{b_0, \dots, b_{i-1}\}}^\perp(b_j))$ the projected block.

Write $\{e_j\}_{j=0}^{p-1}$ for the standard basis of the $\mathbb{Z}^p$ summand (norm 1) and $\{\sqrt{2}f_m\}_{m=0}^{q-1}$ for the $\sqrt{2}\mathbb{Z}^q$ summand (norm $\sqrt{2}$), and $\mathbb{R}^p, \mathbb{R}^q \subset \mathbb{R}^{p+q}$ for the spans of the two families; the algorithm is given only the Gram matrix of an arbitrary basis, so this orthogonal basis exists but is hidden, and finding it is the task. Set $k := p - 1$ for the cut between the two blocks: the cut must satisfy $k < p$ for the front block to be able to terminate as an orthonormal subset of the $\mathbb{Z}^p$ summand (and for Lemma E.2’s Case 1 to find a unit vector outside it), and $k = p - 1$, the largest such cut, keeps both blocks’ SVP dimensions at most p .

E.1 Block reduction for $\mathbb{Z}^d$ and its obstruction

Ducas [Duc23, Thms 1, 2] solves $\mathbb{Z}^d$ -LIP with $\text{poly}(d)$ SVP calls in dimension $\lfloor d/2 \rfloor + 1$, via a 2-block slide reduction with invariant $\text{vol}(B_{[0:k-1]})^2 \in \mathbb{Z}_{>0}$. Its progress lemma is [Duc23, Lemma 4]: if $S \subset \mathbb{Z}^d$ is primitive of rank k with $\text{vol}(S) > 1$, then $\lambda_1(\pi_S^\perp(\mathbb{Z}^d)) \leq \sqrt{1 - 1/d}$. This is the per-round drop of the integer potential described above, for the hypercubic class.

This lemma is false for A' : take $S = \mathbb{Z}^{k-1} \oplus \sqrt{2}\mathbb{Z} \subset A'$ (rank k , $\text{vol}^2 = 2 > 1$). Then $e_{k-1} \in A'$, $e_{k-1} \perp S$, $\|e_{k-1}\| = 1$, so $\lambda_1(\pi_S^\perp(A')) = 1 > \sqrt{1 - 1/(p+q)}$, so the lemma fails.

Nor can the input itself be transformed into a hypercubic class: the determinant class in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ of a rational quadratic form of even rank is unchanged by rescaling, dualizing, and passing to a commensurable lattice, and for A' that class is $2^q \equiv 2$ (q odd) while for every rationally scaled hypercubic lattice it is 1. The fix is to bound the **product** of the primal and dual steps' contributions. The dual lattice $A'^\vee = \mathbb{Z}^p \oplus \frac{1}{\sqrt{2}}\mathbb{Z}^q$ has its *short* vectors in the q -block — where A' has its *long* vectors — so when the primal step stalls, the dual step compensates.

E.2 The integrality lemma

Lemma E.1 (Analogue of [Duc23, Lemma 2]). *For any sublattice $S \subset A'$ of rank $r \geq 1$: $\text{vol}(S)^2 \in \mathbb{Z}_{>0}$. If $\text{vol}(S)^2 = 1$ then $S \subset \mathbb{Z}^p$ and $S \cong \mathbb{Z}^r$.*

Proof. A $\mathbb{Z}$ -basis of S has the form $M = \begin{pmatrix} A \\ C \end{pmatrix}$ with $A \in \mathbb{Z}^{p \times r}$, $C \in \mathbb{Z}^{q \times r}$, and Gram $G_S = M^T D M = A^T A + 2C^T C \in \mathbb{Z}^{r \times r}$, so $\text{vol}(S)^2 = \det G_S \in \mathbb{Z}_{>0}$. Suppose $\det G_S = 1$. Reducing mod 2: $\det(A^T A) \equiv 1 \pmod{2}$, so $\det(A^T A)$ is odd, hence A has full rank r and $A^T A$ is positive-definite. The Minkowski determinant inequality $\det(P + P') \geq \det P$ for P positive-definite and P' positive-semidefinite (equivalently $\det(I + P^{-1/2} P' P^{-1/2}) \geq 1$, since all eigenvalues of the identity plus a positive-semidefinite matrix are ≥ 1) gives $\det(A^T A) \leq \det G_S = 1$. As $\det(A^T A) \in \mathbb{Z}_{>0}$, it equals 1. Then $\det(I + 2(A^T A)^{-1/2} C^T C (A^T A)^{-1/2}) = 1$ with positive-semidefinite inner factor forces $C^T C = 0$, i.e. $C = 0$. So $S \subset \mathbb{Z}^p$ with $\text{vol}(S) = 1$; [Duc23, Lemma 2] applied inside $\mathbb{Z}^p$ gives $S \cong \mathbb{Z}^r$. $\square$

E.3 The Product Progress Lemma

One primal step and the subsequent dual step of [Duc23, Alg. 1] jointly multiply $\text{vol}(B_{[0:k-1]})$ by the product bounded in the following lemma.

Lemma E.2. Let $S \subset A'$ be primitive of rank $k = p - 1$ with $\text{vol}(S) > 1$. Let v be a shortest nonzero vector of $\pi_S^\perp(A')$, let $\tilde{v} \in A'$ be any lift of v , and set $S' := S + \mathbb{Z}\tilde{v}$ (primitive of rank p). Then

$$\lambda_1(\pi_S^\perp(A')) \cdot \lambda_1(S'^\vee) \leq \sqrt{1 - \frac{1}{p}}.$$

Proof. S' is primitive: v is shortest in $\pi_S^\perp(A')$, hence primitive there. The map $\pi_S^\perp : A' \rightarrow \pi_S^\perp(A')$ is a surjective lattice homomorphism with kernel $\text{span}(S) \cap A' = S$ (by primitivity of S), so $S' = (\pi_S^\perp)^{-1}(\mathbb{Z}v)$ and $A'/S' \cong \pi_S^\perp(A')/\mathbb{Z}v$ is torsion-free. Hence $S'^\vee = \pi_{\text{span}(S')}(A'^\vee)$.

Case 1: $\text{span}(S) \not\subset \mathbb{R}^p$. Since $\dim \text{span}(S) = k < p$, some $e_j \notin \text{span}(S)$; then $0 \neq \pi_S^\perp(e_j) \in \pi_S^\perp(A')$ with $\|\pi_S^\perp(e_j)\| \leq 1$, so $\lambda_1(\pi_S^\perp(A')) \leq 1$. As $\text{span}(S') \supset \text{span}(S)$ has a vector with nonzero $\mathbb{R}^q$ -component, some f_m satisfies $\pi_{\text{span}(S')}(f_m) \neq 0$. Since $\frac{1}{\sqrt{2}}f_m \in A'^\vee$, we get $0 \neq \frac{1}{\sqrt{2}}\pi_{\text{span}(S')}(f_m) \in S'^\vee$ with norm $\leq \frac{1}{\sqrt{2}}$. Product $\leq 1 \cdot \frac{1}{\sqrt{2}} \leq \sqrt{1 - 1/p}$ for all $p \geq 2$.

Case 2: $\text{span}(S) \subset \mathbb{R}^p$. Then $S \subset A' \cap \mathbb{R}^p = \mathbb{Z}^p$, and S is primitive in $\mathbb{Z}^p$. [Duc23, Lemma 4] (applied inside $\mathbb{Z}^p$ at rank $k < p$, $\text{vol} > 1$) gives $\lambda_1(\pi_S^\perp(\mathbb{Z}^p)) \leq \sqrt{1 - 1/p}$, and $\pi_S^\perp(\mathbb{Z}^p) \subset \pi_S^\perp(A')$ transfers the bound. For the dual factor: the $p+q$ vectors $\{e_j\} \cup \{\frac{1}{\sqrt{2}}f_m\} \subset A'^\vee$ span $\mathbb{R}^{p+q}$ and each has norm ≤ 1 , so at least one projects nonzero onto the p -dimensional $\text{span}(S')$; that projection witnesses $\lambda_1(S'^\vee) \leq 1$. Product $\leq \sqrt{1 - 1/p} \cdot 1$. $\square$

Remark E.3. In Case 1 the hypothesis $\text{vol}(S) > 1$ is *not* used; the $\sqrt{2}$ -structure supplies the progress. In Case 2 it is used where [Duc23, Lemma 4] needs it. The counterexample from §E.1 falls under Case 1, where the dual step supplies the factor $\frac{1}{\sqrt{2}}$.

E.4 Termination and correctness of Algorithm 1

In Algorithm 1, *primal-SVP reduction* of a projected block inserts a shortest vector of that block at its first position, so that the primal step sets $\|b_k^*\| = \lambda_1(\pi_{\{b_0, \dots, b_{k-1}\}}^\perp(A))$. *Dual-SVP reduction* of $B_{[0:k]}$ applies a unimodular transform to the first $k+1$ basis vectors making $\|b_k^*\|$ as large as possible, namely $1/\lambda_1(\mathcal{L}(B_{[0:k]})^\vee)$ [Duc23]. *Size reduction* translates each basis vector by integer multiples of the preceding ones so that every Gram–Schmidt coefficient $\mu_{ij} := \langle b_i, b_j^* \rangle / \|b_j^*\|^2$ satisfies $|\mu_{ij}| \leq \frac{1}{2}$. *HKZ* (Hermite–Korkine–Zolotarev) *reduction* of a block size-reduces it and SVP-reduces it and, recursively, each of its projected sub-blocks. All SVP calls are in dimension $\leq \max(k+1, p+q-k) = \max(p, q+1) = p = n/2 + 1$. The algorithm reads and updates only the Gram matrix of B ; “outputting a basis” means outputting the unimodular transform from the input basis to it.

Proof of Theorem 5.1. Termination. By Lemma E.1, $W_t := \text{vol}(B_{[0:k-1]})^2 \in \mathbb{Z}_{>0}$ throughout. Let $S_t := \mathcal{L}(B_{[0:k-1]})$ after iteration t ’s dual step; S_t is primitive in A' (as the lattice generated by the first k vectors of a $\mathbb{Z}$ -basis). We account for the drop in the potential W_t across the primal step of iteration t and the dual step of iteration $t+1$. The primal step sets $\|b_k^*\| = \lambda_1(\pi_{S_t}^\perp(A')) =: \alpha$ and $\mathcal{L}(B_{[0:k]}) =: S'_t$ with $\text{vol}(S'_t) = W_t^{1/2}\alpha$. The next dual step sets $\|b_k^*\| = 1/\lambda_1(S_t'^\vee) =: 1/\gamma$, so $W_{t+1}^{1/2} = \text{vol}(S'_t) \cdot \gamma = W_t^{1/2} \cdot \alpha\gamma$. If $W_t > 1$, Lemma E.2 gives $\alpha\gamma \leq \sqrt{1 - 1/p}$, so $W_{t+1} \leq (1 - 1/p)W_t$. From $W_0 \leq (\max_i \|b_i\|)^{2k}$: at most $2pk \ln \max_i \|b_i\|$ iterations.

The dependence on the input basis can be reduced to a doubly logarithmic one. The argument has two phases: first, a recursion proved next brings $\ln W_t$ to at most $1 + p \ln p$; then the factor $1 - 1/p$ of [Lemma E.2](#) brings the potential to 1. In every iteration we additionally have $W_{t+1} \leq p W_t^{1-1/p}$. The primal step's contribution satisfies $\alpha \leq 1$: since $\dim \text{span}(S_t) = k < p$, some e_j lies outside $\text{span}(S_t)$ and projects to a nonzero vector of $\pi_{S_t}^\perp(\Lambda')$ of norm at most 1. For the dual step, $S_t^\vee$ is a lattice of rank p and covolume $1/\text{vol}(S_t) = (W_t^{1/2}\alpha)^{-1}$. The closed cube of side $2 \text{covol}(S_t^\vee)^{1/p}$ has volume $2^p \text{covol}(S_t^\vee)$, so by Minkowski's convex-body theorem it contains a nonzero point of $S_t^\vee$, of Euclidean norm at most $\sqrt{p} \text{covol}(S_t^\vee)^{1/p}$. This gives $\gamma^2 = \lambda_1(S_t^\vee)^2 \leq p (W_t^{1/2}\alpha)^{-2/p}$. Hence $W_{t+1} = W_t(\alpha\gamma)^2 \leq p \alpha^{2(1-1/p)} W_t^{1-1/p} \leq p W_t^{1-1/p}$. Writing $v_t := \ln W_t$, this reads $v_{t+1} \leq \ln p + (1-1/p) v_t$. Hence $v_t \leq (1-1/p)^t v_0 + p \ln p$, and $v_t \leq 1 + p \ln p$ within $\lceil p \ln v_0 \rceil$ iterations. The factor $1 - 1/p$ of [Lemma E.2](#) then brings the integer potential to 1 within $p(1 + p \ln p)$ further iterations. With $v_0 = \ln W_0 \leq 2k \ln \max_i \|b_i\|$ this gives the bound stated in the theorem (the $+3p$ there absorbs the ceiling and the unpaired first dual step).

Correctness. At termination $\text{vol}(B_{[0:k-1]})^2 = 1$, so by [Lemma E.1](#) the front block $S := \mathcal{L}(B_{[0:k-1]})$ is a rank- k sublattice of $\mathbb{Z}^p$ with an orthonormal basis; since the only vectors of $\mathbb{Z}^p$ of norm 1 are the $\pm e_j$, S is the span of k of the p unit vectors. The final step HKZ-reduces the blocks $B_{[0:k-1]}$ (rank k) and $B_{[k:p+q-1]}$ (rank $q+1$) and size-reduces the full basis, using SVP calls in dimension at most $\max(k, q+1) = n/2 < p$.

For the front block: in an HKZ-reduced basis of S every Gram-Schmidt vector is a unit vector $\pm e_j$ of $\mathbb{Z}^p$ (inductively: the first basis vector is a norm-1 vector of $S \subseteq \mathbb{Z}^p$, hence some $\pm e_j$; projecting S orthogonally to it leaves the span of the remaining unit vectors, whose norm-1 vectors are again $\pm e_j$); the size-reduction coefficients $\mu_{ij} = \langle b_i, b_j^* \rangle / \|b_j^*\|^2 = \langle b_i, \pm e_j \rangle$ are integers, hence are reduced to 0; the front block becomes an orthonormal set of k unit vectors.

For the back block: $\pi_S^\perp(\Lambda') = \mathbb{Z}^{p-k} \oplus \sqrt{2}\mathbb{Z}^q = \mathbb{Z} \oplus \sqrt{2}\mathbb{Z}^q$. Its shortest vector is the remaining unit vector, unique up to sign (everything outside $\mathbb{Z}e$ has norm $\geq \sqrt{2}$), so the first HKZ vector of the back block is $\pm e$; the projection orthogonal to it is $\sqrt{2}\mathbb{Z}^q$, whose HKZ basis is its orthogonal basis by the same argument as for the front block (scaled by $\sqrt{2}$). Size reduction again clears the cross terms: against a unit vector $b_j^* = \pm e$ the coefficient is an integer, and against a vector $b_j^* = \pm\sqrt{2}f$ the coefficient is $\langle b_i, \sqrt{2}f \rangle / 2 \in (2\mathbb{Z})/2 = \mathbb{Z}$. The output basis is therefore the hidden orthogonal basis $(e_0, \dots, e_{p-1}, \sqrt{2}f_0, \dots, \sqrt{2}f_{q-1})$ of Λ' up to signed permutation. (These terminal HKZ reductions use SVP dimension $n/2$ rather than the $\lfloor n/4 \rfloor + 1$ that a recursive half-dimension LIP solve would give; the main loop already requires dimension $p = n/2 + 1$, so the theorem's bound is unaffected, and plain HKZ is also how [\[Duc23, Alg. 1\]](#) finishes for $\mathbb{Z}^d$.) $\square$

Corollary E.4. $[\mathbb{Z}^p \oplus \sqrt{s}\mathbb{Z}^q]\text{-LIP} \leq_{\det} \text{poly}(p) \times \text{SVP}(p)$ (*deterministic reduction*) for any integer $s \geq 2$ and $p > q \geq 0$. (Replace “mod 2” by “mod s ” in

[Lemma E.1](#); [Lemma E.2](#) Case 1 gives product $\leq 1/\sqrt{s} \leq \sqrt{1-1/p}$ for $p \geq 2$; Case 2 is unchanged.)

Remark E.5 (relation to [BN24]). Bambury–Nguyen [BN24, Thm 1] prove that for any rank- d lattice $\Lambda \subseteq \mathbb{Z}^d$ with $\lambda_1(\Lambda)\lambda_1(\Lambda^\vee) < 1 - \varepsilon$, a nonzero vector of norm $\leq \lambda_1(\Lambda)$ in Λ or one of norm $\leq \lambda_1(\Lambda^\vee)$ in $\Lambda^\vee$ can be found with polynomially many SVP calls in dimension $\lfloor d/2 \rfloor + 1$, terminating as soon as one is found; their Theorem 4 returns a full orthogonal basis of $\Lambda \cong \mathbb{Z}^d$ with a γ -approximate oracle for any $\gamma < \sqrt{2 - 2/d}$. The class $[\Lambda']$ satisfies the hypothesis of their Theorem 1 with $\lambda_1(\Lambda')\lambda_1(\Lambda'^\vee) = 1/\sqrt{2}$, and [Theorem 5.1](#) can be read as a near-hypercubic instance of their framework that returns all shortest vectors. The analysis here builds on theirs: [BN24, Lemma 1] expresses the per-round volume drop as the product of the primal and dual steps’ contributions, and their Theorem 1 bounds that product by $\lambda_1(\Lambda)\lambda_1(\Lambda^\vee)$ while neither half-block contains a shortest vector. [Lemma E.2](#) bounds the same product under $\text{vol}(S) > 1$ alone, using the structure of $[\Lambda']$, so that the loop runs until the front-block volume reaches 1, where all shortest vectors can be read off (the direct analogue of [Duc23, Lemma 4] fails for Λ' , §E.1). An alternative route iterates [BN24, Thm 1], removing each found primal or dual shortest vector and recursing; the residual lattices remain of the form $\mathbb{Z}^{p'} \oplus \sqrt{2}\mathbb{Z}^{q'}$ until $q' = 0$, where the hypercubic solver of [Duc23] applies. We organize the argument as a single termination bound.

F Falcon and NTRU-Type Schemes Are Not Affected

Falcon [PFH20] uses the same field $K_n = \mathbb{Q}(\zeta_{2^e})$, so τ exists. However, the τ -cocycle construction does not transfer, for two independent structural reasons. Throughout this section we use Falcon’s standard notation: q denotes the NTRU modulus (not the integer $q = n/2 - 1$ of [Corollary 4.6](#)), f, g, F, G denote the four ring elements of Falcon’s secret basis (unrelated to the field F of §§3–4), and σ_{sk} is the Gaussian width of the NTRU secret.

(i) **No public Hermitian Gram.** Falcon publishes $h = gf^{-1} \bmod q \in R_n/q$ [PFH20, eq. (3.17)], not the Gram of the secret basis B . Two secret keys with the same h have different Grams (they differ by left-multiplication by an element of $\text{GL}_2(R_n)$ that need not be unitary), so constraint (C2) of §4.1 — which has coefficients depending on the Gram matrix — is not a function of the public key. Without (C2), the lattice of matrices satisfying (C1) alone has rank $2n$, not n , and offers no advantage.

(ii) $\det B = q \neq 1$. [Lemma 4.2](#)’s integral isometry requires $B \in \text{SL}_2(R_n)$, and [Lemma 4.4](#)’s minimum $n/4$ requires $\det V_\tau = 1$. Falcon’s secret basis is $B = \begin{pmatrix} g & -f \\ F & -G \end{pmatrix}$ [PFH20, eq. (2.2)], and the NTRU equation $fG - gF = q$ [PFH20, eq. (3.15)] is $\det B = q \notin R_n^\times$; the resulting cocycle $qB^{-1}\tau(B)$ has determinant q^2 and $Q^{(\tau)}$ -value $q^2 \cdot n/4$, above the Gaussian heuristic (the expected first minimum of a random lattice of the same volume) of any publicly derived q -ary lattice (one containing $qM_2(R_n)$) — a *worse* target than $(f, g) \in A_h := \{(u, v) \in R_n^2 : uh \equiv v \bmod q\}$ directly.

More generally, consider any polynomial in the entries of B and their Galois conjugates whose norm is independent of σ_{sk} . Each monomial mixes short factors (f, g , of size $\sim \sigma_{\text{sk}}$) with long ones (the completion F, G , of size $\sim q/\sigma_{\text{sk}}$); for the product to be σ_{sk} -independent the two must balance, and any non-constant balanced monomial has norm $\geq \Theta(q)$. Reason (i) and this scale-imbalance argument apply to any NTRU-type scheme, not only to Falcon. The distribution-independent minimum of [Lemma 4.4](#) is possible for HAWK because all four entries of B have the same scale. To the best of our effort, none of the related side-constructions — descent via N_{K_n/K_n^τ} or $N_{K_n/F}$, or the LIP reduction of [Corollary E.4](#) applied to q -ary lattices — helps against Falcon: each either reduces to the standard subfield attacks [\[ABD16; KF17\]](#) or has no Falcon instance.